

Replacement and Reputation^{*}

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Abstract

Does electoral replacement ensure that officeholders eventually act in voters' interests? We study a reputational model of accountability. Voters observe incumbents' performance and decide whether to replace them. Politicians may be “good” types who always exert effort or opportunists who may shirk. We find that good long-run outcomes are always attainable, though the mechanism and its robustness depend on economic conditions. In environments conducive to incentive provision, some equilibria feature sustained effort, yet others exhibit some long-run shirking. In the complementary case, opportunists are never fully disciplined, but selection dominates: every equilibrium eventually settles on a good politician, yielding permanent effort.

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1. Introduction

In various delegated decision-making settings, a principal’s ultimate tool is the power to replace an agent. A firm’s board can fire a poorly performing CEO; patients can switch health-care providers after bad service; and ineffective bureaucrats or organizational leaders may be dismissed. But the most prominent context is democratic politics, where the primary instrument voters use to control officeholders is re-election. An influential body of work, starting with Barro (1973) and Ferejohn (1986), has studied the consequences of voters’ authority to replace incumbents. Such authority is essential not only to discipline politicians’ behavior, but also to select good politicians (Fearon, 1999; Besley, 2005).

Indeed, much further back, Mill (1859) stressed the importance of the *long-run* implications of selection, writing that “The worth of a State, in the long run, is the worth of the individuals composing it.” Yet we know surprisingly little from existing formal analyses about the long-run consequences of replacement in environments with both adverse selection and moral hazard. Much of the theoretical political-accountability literature (elaborated subsequently) analyzes models with short horizons or with short term limits.¹ A fundamental question thus remains: absent other frictions, does the replacement mechanism ensure that eventually voters can identify good politicians, and/or that officeholders will act in voters’ interests?

This paper answers that question in a simple and stylized model that we view as a natural benchmark. Our model takes inspiration from not only prior work on accountability, but also the reputation literature (e.g., Fudenberg and Levine, 1992). While the model is applicable more broadly, for concreteness we filter it through a political lens.

A pool of long-lived politicians interacts with a sequence of short-lived (or myopic) voters. In each period, the current voter observes the incumbent’s past performance and decides whether to retain him or replace him with a fresh draw from the politician pool. The officeholder—old or new—then chooses, covertly, whether to exert effort (work or shirk), which maps stochastically into performance. Politicians are one of two types: a *good* type who always exerts effort, or an *opportunistic* type who values office but dislikes effort. Voters simply want effort;² but this means they instrumentally benefit from good types, and politicians have an incentive to build a reputation. Apart from the retention decision, voters

¹This contrasts with an empirical literature on the considerable length of political careers. See, for example, structural estimates by Diermeier, Keane, and Merlo (2005) and a study on legislative effectiveness by Padró-i Miquel and Snyder Jr. (2006), both highlighting how performance and retention of officeholders evolve significantly over long tenures.

²They may also incur a cost of replacing incumbents, but we suppress that for this introduction.

have no recourse to any other incentive instruments such as transfers.

In this setting, we obtain a sharp but nuanced answer to the motivating question of long-run outcomes. On the one hand, there is always at least one equilibrium that eventually has sustained effort from incumbents.³ On the other hand, whether this good long-run outcome arises in *every* equilibrium depends on the economic environment. Intuitively, the environment is conducive to incentive provision if effort costs are low, monitoring (i.e., how performance reflects effort) is precise, and politicians are patient. Then there is an equilibrium in which even opportunistic incumbents exert effort in every period (even in the short run). But we show that in this case, there also exist equilibria in which, even in the long run, opportunistic politicians will periodically hold office and shirk. We identify a novel feature of all such equilibria that do not deliver good long-run outcomes: *replacement despite a favorable reputation*. That is, voters sometimes replace an incumbent even when his past performance indicates that he is more likely to be a good type than the average politician. Voters do so when they (correctly) anticipate that an opportunistic incumbent with the favorable reputation is sufficiently more likely to shirk than a newly-installed opportunist. The upshot is that in environments conducive to incentive provision, equilibrium selection—which can be viewed as political norms—crucially matter for outcomes, even in the long run.

Now consider the complementary case when parameters are not conducive to incentive provision: there is no equilibrium that sustains effort in every period from opportunists. Paradoxically perhaps, this turns out to guarantee effort in the long run. The mechanism operates through selection. We show that now, in every equilibrium, every opportunistic officeholder is eventually replaced, whereas eventually some good-type incumbent will be retained forever. The logic assuring long-run selection is subtle. To appreciate why, suppose good types are scarce. We establish that the value of a newly-installed politician (a voter’s endogenous “outside option”) is then *arbitrarily low* across all equilibria, meaning that newly-installed officeholders almost certainly shirk. That makes voters reluctant to replace any incumbent. However, it turns out that opportunists are always driven to histories at which voters believe the incumbent is overwhelmingly likely to be an opportunist who will shirk—so much so that the low outside option is preferable. Conversely, every good type has a chance of reaching and then maintaining a high enough reputation to avoid replacement.

The dichotomy in the two previous paragraphs is summarized in our main result, [Theo-](#)

³ As elaborated in [Section 2](#), our equilibrium notion is that of “personal symmetric” weak perfect Bayesian equilibrium. Roughly speaking, “personal” means that the interaction between every incumbent and the voters is self-contained, and “symmetric” means it is ex-ante identical across all the politicians.

rem 1: there is no equilibrium that has “full effort” (incumbents exert effort in every period) if and only if all equilibria have “eventual full effort” (after some point, incumbents exert effort in all subsequent periods). The result crystallizes a sharp tradeoff between the *possibility* of strong discipline (i.e., even in the short run) and the *guarantee* of either discipline or selection in the long run.

Related Literature. Our paper contributes to and sits at the intersection of two literatures: political accountability and reputation formation.

In the accountability literature, the idea that replacement is an instrument for selection, rather than just controlling moral hazard, was popularized by [Fearon \(1999\)](#).⁴ Much of the literature studying selection, surveyed by [Ashworth \(2012\)](#) and [Duggan and Martinelli \(2017\)](#), uses models with either just two periods or a two-term limit; exceptions include [Banks and Sundaram \(1993\)](#), [Duggan \(2000\)](#), [Bernhardt, Dubey, and Hughson \(2004\)](#), and [Schwabe \(2009\)](#). None of these papers, nor virtually any others, have results on selection leading to voter-optimal outcomes in the long run—in any equilibrium, much less all equilibria, as we find under some conditions. The only exception we are aware of is [Anesi and Buisseret \(2022\)](#).⁵ They construct one equilibrium of their model that is approximately first best when each politician’s type follows an irreducible Markov chain and all agents—politicians and the voter—are almost fully patient. By contrast, we have fully-persistent politician types with fixed discounting, and short-lived (or impatient) voters. [Anesi and Buisseret’s \(2022\)](#) construction relies on patient voters using sophisticated punishment schemes that are themselves supported by “grim-trigger” strategies; in our setting, it is instead the very inability to discipline opportunists that guarantees eventual positive selection.

When our environment is conducive to incentive provision, our construction of an equilibrium in which incumbents shirk even in the long run uncovers a new mechanism for inefficiency. It owes to the dynamics of opportunistic incumbents’ effort over their careers, and

⁴ We will not attempt to discuss the large literature on accountability with just moral hazard; see [Duggan and Martinelli \(2017\)](#) for a survey.

⁵ In their extension that incorporates adverse selection, [Acharya, Lipnowski, and Ramos \(2025\)](#) study a model with an opportunistic type and a bad type who always shirks. They show that a voter-optimal equilibrium weeds bad politicians out from office; yet, because opportunistic politicians still face moral hazard, equilibria need not yield sustained effort in the long run. We discuss adding bad types to our model in [Section 4](#). Another paper concerned with long-run outcomes is [Duggan and Forand \(2025\)](#); they assume that an incumbent’s type is publicly revealed after he assumes office, similar to one section of [Kartik and Van Weelden \(2019\)](#) that does not have term limits but assumes types are revealed after two periods in office. Such exogenous type revelation limits (in the latter paper) or even eliminates (in the former) the reputation-building horizon. [Kartik and Van Weelden \(2019\)](#) focus on how their short-horizon reputation-building affects incumbency advantage. [Duggan and Forand \(2025\)](#) focus on how incumbents can use current policies to manipulate future states and policies, and how patient voters may nevertheless attain good outcomes.

the resulting replacement despite a favorable reputation described earlier.⁶ In [Section 3.3](#) we explain why the mechanism is distinct from that of the inefficient equilibrium of [Myerson \(2006\)](#), which owes to large replacement costs. The coexistence of good and bad equilibria in our incentive-conducive environments may also be reminiscent of folk theorems in repeated complete-information games (e.g., [Fudenberg and Maskin, 1986](#)). However, we have short-lived voters who may learn about politician types (so there is incomplete information), which makes the reputation literature discussed next—with quite different prior results—the more relevant comparison.

Most models in the reputation literature have a *fixed* long-lived player interacting with a sequence of short-lived players. [Fudenberg and Levine \(1989, 1992\)](#) derive tight bounds for the long-lived player’s equilibrium payoffs as he becomes fully patient. [Cripps, Mailath, and Samuelson \(2004\)](#) show that when monitoring is imperfect, although the [Fudenberg and Levine \(1992\)](#) bounds apply *ex ante*, the long-lived player will eventually lose his reputation in the sense that his opponents will learn his type.⁷ By contrast, our model has multiple long-lived players, and our focus is on their endogenous replacement. The possibility of replacement implies that the short-lived players may not learn any given long-lived player’s type, and because the replacement probabilities are endogenous, the long-lived players’ effective discount factors are endogenous and evolving, possibly bounded away from full patience. We instead focus on outcomes from the short-lived players’ vantage, which we view as appropriate for many applications, including electoral accountability.

There is prior work on reputation models with competition between long-lived players, notably [Hörner \(2002\)](#) and, more recently, [Deb and Fanning \(2025\)](#). Besides other differences (e.g., in the behavior of committed types and, in the former paper, endogenous pricing), the nature of the short-lived players’ “outside option” is distinct. The central tradeoff our voters face is between continuing interaction and restarting it afresh: they compare the anticipated behavior of a retained incumbent with that of a new politician. Both are endogenous, but the latter is an equilibrium constant. By contrast, in the aforementioned competitive models, the consumers’ outside option is the equilibrium value of switching to a rival firm (whether

⁶One can contrast this equilibrium feature with the literature on managerial turnover with dynamic contracting; see, for example, [Garrett and Pavan \(2012\)](#) and references therein. Papers in that literature characterize *ex-ante* optimal retention and compensation policies for a long-lived principal with commitment power. Optimal contracts often exhibit “entrenchment” in the sense that retention standards become more lenient over time to reduce information rents. Such entrenchment also features in certain analyses of accountability, like [Acharya, Lipnowski, and Ramos \(2025\)](#).

⁷A number of subsequent papers propose models in which reputation can be sustained in the long run. The mechanisms include competition between multiple long-lived players (e.g., [Hörner, 2002](#); [Deb and Fanning, 2025](#)), the long-lived player’s type changing over time (e.g., [Ekmekci, Gossner, and Wilson, 2012](#)), and the short-lived players having limited memories (e.g., [Liu and Skrzypacz, 2014](#); [Pei, 2024](#)).

a previously established firm or a new entrant), which evolves with the history and produces distinct forces from simply restarting afresh.⁸

Finally, ongoing work by Ali (2025) studies a model related to ours. He has only two politicians (among whom voters may rotate), and studies a class of equilibria with a “pure Markovian” structure. While Ali (2025) also finds conditions under which all his equilibria have eventual full effort, the mechanism does not operate through selection; to the contrary, a full-effort equilibrium exists under his conditions. Section 4 elaborates.

2. Model

Setting. We consider an infinite-horizon game in discrete time, with periods indexed by $t \in \mathbb{Z}_{\geq 0} \equiv \{0, 1, 2, \dots\}$. There is a pool of ex-ante identical long-lived players, whom we refer to as *politicians*. Each period, only one long-lived player, whom we call the *incumbent*, is active. An incumbent remains active until he is replaced, after which he is never active again. There is also an infinite sequence of short-lived players, whom we call *voters*. The voters arrive one in each period, and each voter plays the game only in the period they arrive.

In each period other than the initial period, the current voter decides whether to *keep* the previous period’s incumbent or *replace* him with a new politician. In the initial period 0 (there being no prior incumbent), the voter makes no choice. (We will typically suppress the qualifier “current” hereafter.) Then, the incumbent chooses an action $a \in A := \{0, 1\}$. We refer to $a = 0$ as *shirking* and $a = 1$ as, interchangeably, *working* or *exerting effort*. This action stochastically generates a public signal s drawn from a finite set S according to a monitoring structure $f : A \rightarrow \Delta S$. Let $f_a(s)$ denote the probability of signal s conditional on action $a \in A$, and let $f_{\tilde{a}}(s) := \tilde{a}f_1(s) + (1 - \tilde{a})f_0(s)$ denote the aggregate probability of signal s if the incumbent exerts effort with probability $\tilde{a} \in [0, 1]$.

When a politician is active, his stage-game payoff from action a is $u_P(a)$. Once a politician is replaced, he is never again active and receives a fixed payoff, which we normalize to 0. The voter’s payoff from action a is $u_V(a)$, and they further incur an additive cost $c \geq 0$ if they replace the incumbent.⁹ We assume that the voter’s preferred action is different from the incumbent politician’s (myopically) favorite action, and that a politician’s least favorite

⁸ Naturally, our tradeoff is not present either in models in which the long-lived player exits according to some exogenous process (Mailath and Samuelson, 2001; Tadelis, 2002; Ekmekci, Gossner, and Wilson, 2012).

⁹ We allow for a positive replacement cost to clarify that our analysis is not driven by excessive voter indifference, e.g., being indifferent to replacement when they expect the same incumbent behavior after retention or replacement. We will be interested in the case where the replacement cost is not too large.

outcome is to be replaced. Without further loss of generality, we normalize payoffs so that $u_V(a) = a$ and $u_P(a) = 1 - \kappa a$ for some $\kappa \in (0, 1)$. We also assume the monitoring structure is informative, that is, $f_0(\cdot) \neq f_1(\cdot)$. Finally, assume no signal perfectly reveals the politician to be shirking, that is, $f_1(s) > 0$ for every $s \in S$.

Following [Besley \(2006\)](#) and [Besley and Smart \(2007\)](#), among others, we take seriously that some politicians are intrinsically motivated. With probability $\pi_0 \in (0, 1)$, a given politician is a *good type* who always plays $a = 1$ when active. With complementary probability $1 - \pi_0$, he is an *opportunistic type* who maximizes the expectation of his discounted average payoff $\sum_{t=\tau_0}^{\tau_1-1} (1 - \delta)\delta^{t-\tau_0} u_P(a_t)$ where a_t is his period- t action, τ_0 is the period that he becomes active, τ_1 is the period that he is replaced (which we label as ∞ if he is never replaced), and $\delta \in (0, 1)$ is his discount factor. Each politician's type is his private information, and types are independent. Hence, after a voter replaces an incumbent, a newly drawn politician is the good type with probability π_0 . We assume that the replacement cost (which can be zero) is small relative to this uncertainty: $c < \min\{\pi_0, 1 - \pi_0\}$.

Interpretations. An incumbent's choice of effort can be interpreted broadly to capture various familiar conflicts of interest in political agency. Working ($a = 1$) can correspond to costly activities that benefit the electorate, such as the effective provision of public goods. Conversely, shirking ($a = 0$) can represent rent-seeking behavior, such as engaging in corruption or implementing inefficient pork-barrel projects that benefit special interests.

We model voters as short lived for two reasons. First, it is a way to capture an electorate composed of many long-lived but uncoordinated voters. Rather than a mass electorate coordinating on complex strategies guided by forward-looking considerations, we take the view that each voter simply votes for the alternative that maximizes their payoff in the current period. Such myopic behavior—which can also be viewed as an equilibrium refinement with many voters, none of whom is pivotal—is mathematically equivalent to a sequence of short-lived voters. Second, we are ruling out sophisticated forward-looking coordination schemes between politicians and voters. Ultimately, as in the canonical reputation literature, our assumption sharpens the focus on how opportunistic politicians' career concerns, rather than voters' long-term considerations, shapes outcomes. We return to this issue in [Section 4](#).

Strategies and equilibria. We will focus on equilibria in which players' behavior depends only on the current officeholder's past outcomes. To that end, the (*public*) *career history* of a given incumbent records the sequence of public signals he has generated in the past.

Let $\mathcal{H} := S^{<\infty} = \bigsqcup_{t=0}^{\infty} S^t$ denote the set of career histories.¹⁰ Note that $\mathcal{H} \setminus \{\emptyset\} = \mathcal{H} \times S$. Our solution concept entails a politician strategy $\sigma_P : \mathcal{H} \rightarrow [0, 1]$, a voter strategy $\sigma_V : \mathcal{H} \setminus \{\emptyset\} \rightarrow [0, 1]$, and a voter belief map $\pi : \mathcal{H} \rightarrow [0, 1]$. Here, $\sigma_P(h)$ denotes the probability that an opportunistic incumbent works when his career history is h ; $\sigma_V(h)$ denotes the probability the current voter replaces an officeholder whose career history is $h \neq \emptyset$; and $\pi(h)$ is the probability voters assign to a politician being the good type (his *reputation*) when his career history is h . Given such a triple, it is useful to record the voter-expected incumbent efforts and the opportunistic incumbents' continuation values. Formally, suppressing their dependence on $(\sigma_P, \sigma_V, \pi)$ for brevity, we define $\alpha : \mathcal{H} \rightarrow [0, 1]$ via

$$\alpha(h) := \pi(h) + [1 - \pi(h)] \sigma_P(h),$$

and $v : \mathcal{H} \rightarrow [0, 1]$ as the unique solution to the recursive equation

$$v(h) = (1 - \delta) [1 - \kappa \sigma_P(h)] + \delta \sum_{s \in S} f_{\sigma_P(h)}(s) [1 - \sigma_V(h, s)] v(h, s), \quad \forall h \in \mathcal{H}.$$

We call a triple $(\sigma_P, \sigma_V, \pi)$ a **personal symmetric weak perfect Bayesian equilibrium** (hereafter **equilibrium**) if it satisfies the following three properties:

1. Any $h \in \mathcal{H}$ has

$$\sigma_P(h) \in \arg \max_{a \in [0, 1]} \left\{ (1 - \delta)(1 - \kappa a) + \delta \sum_{s \in S} f_a(s) [1 - \sigma_V(h, s)] v(h, s) \right\}.$$

2. Any $h \in \mathcal{H} \setminus \{\emptyset\}$ has $\sigma_V(h) \in \arg \max_{x \in [0, 1]} \{(1 - x)\alpha(h) + x[\alpha(\emptyset) - c]\}$.
3. We have $\pi(\emptyset) = \pi_0$, and any $h \in \mathcal{H}$ and $s \in S$ with either $h = \emptyset$ or $\sigma_V(h) < 1$ has

$$f_{\alpha(h)}(s)\pi(h, s) = \pi(h)f_1(s).$$

The first equilibrium condition is a recursive formulation of sequential rationality for incumbents. The second condition is for voters: being short-lived, they make replacement decisions that maximize the current effort they expect from an officeholder net of replacement cost. The third condition incorporates voters' Bayesian updating: it requires that (i) any newly-installed incumbent has reputation π_0 , the prior probability of a good type; and (ii) reputation at any career history (h, s) is derived from Bayes' rule applied to h so long as

¹⁰ The full public history—the full history available to voters—consists of an incumbent career history, all past replacement decisions, and the career histories associated with all previously replaced politicians.

either $h = \emptyset$ (so s is the first signal generated by the incumbent) or the incumbent has been retained with positive probability at h .¹¹

Following Banks and Sundaram’s (1993) terminology, the “personal” nomenclature in our equilibrium concept reflects that the interaction between any officeholder and the voters depends only on the sequence of signals generated by that politician—not, for example, on the calendar time at which the politician took office nor on what transpired prior to when he took office.¹² The “symmetric” adjective reflects that all opportunistic officeholders respond to their career history in the same way and are treated identically by voters. Combined, these two properties reflect the premise that the role of replacement is to start the interaction with a new politician afresh; put differently, any equilibrium has a single (endogenously-determined) “outside option”—a voter’s value from replacement. This approach is standard in the political-economy literature; besides Banks and Sundaram (1993), see, for example, Duggan (2000) or Van Weelden (2013). Moreover, as we shall see, it does not preclude rich dynamics within any incumbent’s career.

As a preliminary step, we ensure that these refinements do not preclude existence.

Proposition 1. *An equilibrium exists.*

The proof in Appendix A.1 constructs an auxiliary game whose perfect Bayesian equilibria (PBE) are equilibria of our game. The auxiliary game has only one politician. It mirrors the interaction in our game between a given politician and the voters, modifying it in two ways. First, the politician’s first-period choice is $\tilde{a} \in [0, 1]$, representing the probability with which he chooses action 1; in all other periods, the politician makes a choice from $A \equiv \{0, 1\}$. While the politician can mix in subsequent periods, we require him to not mix (over the action set $[0, 1]$) at the initial history. Second, the voter’s payoff when they replace the politician is $\pi_0 + \tilde{a}(1 - \pi_0) - c$. Intuitively, our specification of the voter’s payoff upon firing the politician captures the symmetry requirement in the original game. We require the politician not to mix in the first period to avoid voters updating beliefs as play progresses about their payoff from replacement (contrary to symmetry in the original game). Making the politician’s first-period choice continuous secures the existence of such PBE in the auxiliary game, following standard arguments à la Kreps and Wilson (1982) and Fudenberg and Levine (1983).

¹¹ We refer to our solution concept as “weak” PBE because beliefs can be arbitrary at certain non-initial career histories that are off path—specifically, at any (h, s) with $h \neq \emptyset$ and $\sigma_V(h) = 1$. It is still stronger than textbook definitions of weak PBE that impose no belief restrictions off path (Mas-Colell, Whinston, and Green, 1995, Definition 9.C.3, p. 285). Section 3.1 elaborates on our concept vs. “full” PBE.

¹² We have also imposed that the politician conditions only on his own *public* career history, not on his own past actions. This restriction is for notational simplicity: Because the game has “product monitoring”, standard arguments show this feature is without loss in terms of outcome equivalence.

3. Results

Our goal is to study whether replacement ensures that incumbents exert effort. To that end, we introduce two criteria. Using bold font for random variables, let $\mathbf{a}_t \in \{0, 1\}$ indicate the incumbent's chosen effort in period t , which will in particular be equal to 1 if the incumbent is the good type. We say that an equilibrium **attains full effort** if $\mathbf{a}_t = 1$ almost surely for every $t \in \mathbb{Z}_{\geq 0}$, and it **attains eventual full effort** if there almost surely exists some $\tau \in \mathbb{Z}_{\geq 0}$ such that $\mathbf{a}_t = 1$ for all $t \geq \tau$. In words, an equilibrium attains full effort (FE) if incumbents always work (on the path of play), and it attains eventual full effort (EFE) if, eventually, incumbents work in every period.¹³ The distinction between FE and EFE is thus one of timing; an equilibrium with FE also has EFE, but the converse is not generally true. In addition, one can also draw a distinction between whether, for any given parameters, either property holds in *no* equilibrium, *some* equilibrium, or *all* equilibria.

A strong desideratum is that all equilibria attain FE; conversely, it would be troubling if no equilibrium attains even EFE. We will see that neither of these is possible, no matter the parameters.¹⁴ Instead, our main result is that parameters divide sharply into two cases: either there is an equilibrium that attains FE, but then there is also an equilibrium that fails even EFE; or there is no equilibrium that attains FE, but then all equilibria attain EFE. The following table summarizes.

	Effort always (FE)	Effort eventually (EFE)
Condition FEI	<i>Some</i> equilibrium	<i>Some</i> equilibrium
$\neg$ Condition FEI	<i>No</i> equilibrium	<i>All</i> equilibria

Table 1: The equilibrium possibilities. Full Effort Incentives (FEI) is a condition on parameters.

The table's [Condition FEI](#) on parameters is the following:

Condition FEI (*Full Effort Incentives*). *Some $v : S \rightarrow \mathbb{R}_{\geq 0}$ exists such that*

$$(1 - \delta)(1 - \kappa) + \delta \sum_{s \in S} f_1(s)v(s) \geq (1 - \delta)1 + \delta \sum_{s \in S} f_0(s)v(s) \quad (\text{IC}_{\text{FEI}})$$

¹³ One might be interested in weaker variants of both FE and EFE. Not only would our positive results obviously extend, but, as elaborated in Subsections 3.2 and 3.3, so do our negative results.

¹⁴ One can show that absent good types, no equilibrium attains EFE when [Condition FEI](#) below fails. Even with a good type, no equilibrium attains EFE in the canonical noisy-monitoring product-choice game between a long-lived firm and short-lived consumers when there is no possibility of replacement ([Cripps, Mailath, and Samuelson, 2004](#)). In a different model, [Pei \(2024\)](#) presents a result in the vein of all equilibria attaining FE.

and

$$(1 - \delta)(1 - \kappa) + \delta \sum_{s \in S} f_1(s)v(s) \geq \max_{s \in S} v(s). \quad (\text{PK}_{\text{FEI}})$$

Condition FEI is independent of politicians' initial reputation π_0 and voters' cost of replacement c because the condition stems from contemplating an auxiliary game without good types and without a replacement cost. In the auxiliary game, if all incumbents always exert effort, then voter incentives are trivially satisfied. Standard recursive arguments then imply that under some conditions, there are equilibria in which appropriate voter behavior can incentivize incumbents to always exert effort. The associated incentive-compatibility and promise-keeping constraints are precisely those in **Condition FEI**, inequalities (IC_{FEI}) and (PK_{FEI}) respectively. **Condition FEI** is satisfied if and only if the politicians' discount factor δ is high enough, their effort cost κ is small enough (recall we have normalized their benefits of holding office to 1), and the monitoring structure f is sufficiently informative.¹⁵ Parameters that satisfy **Condition FEI** thus correspond to environments that are conducive to incentive provision.

3.1. Main Result

We now present our main result.

Theorem 1. *The following are equivalent:*

1. *All equilibria attain eventual full effort;*
2. *No equilibrium attains full effort;*
3. *Condition FEI fails.*

An implication is that no matter the parameters, there is always at least one equilibrium that attains EFE, but also at least one equilibrium without FE.

We view the main economic lesson from the theorem as the sharp contrast summarized in **Table 1**: environments conducive to incentive provision yield the *possibility* of FE; those

¹⁵ We emphasize that **Condition FEI** is a joint condition on the parameters (δ, κ, f) . So, for instance, the required discount factor is higher when the effort cost is higher or the monitoring structure is less informative in the **Blackwell (1953)** sense. As an illustration, with binary signals of precision $p \in (1/2, 1)$ (that is, $S = \{0, 1\}$ and $f_a(a) = p$), it can be verified using **Lemma 12**—which shows that, no matter the monitoring structure, there no loss in restricting to $v(s) \in \{0, \bar{v}\}$ for some $\bar{v} > 0$ —that **Condition FEI** reduces to

$$\delta \geq \frac{\kappa}{p - (1 - p)(1 - \kappa)},$$

where the right-hand side is decreasing in p and increasing in κ .

that are not conducive *guarantee* EFE. More precisely, when [Condition FEI](#) holds, although there are equilibria that attain FE, there are also equilibria that fail EFE. By contrast, when [Condition FEI](#) fails, there is no FE equilibrium, but perpetual effort is inevitable after finite time in every equilibrium. As we will elaborate, EFE is guaranteed through (eventual) selection of a good type; paradoxically, when [Condition FEI](#) holds, the very possibility of providing strong incentives undermines assuring good selection or assuring good long-run outcomes. To our knowledge, this is a novel tradeoff between discipline and selection.

The proof of [Theorem 1](#) is in [Appendix A.2](#). The remainder of this section discusses the intuition and forces underlying the theorem. But before that, we comment on an aspect of using (personal, symmetric) *weak* PBE as our solution concept. Condition (3) of the equilibrium definition allows for arbitrary beliefs off path immediately after a voter retains an incumbent who should have been certainly replaced. Dropping the condition’s qualification that $\sigma_V(h) < 1$ would impose belief requirements off path that yield (personal, symmetric) PBE, without the “weak” moniker. [Appendix A.1](#) assures existence for this solution concept as well. Our equilibrium constructions under [Condition FEI](#) benefit from the off-path latitude allowed by weak PBE, as flagged in [Appendix A.2](#). At the same time, the weaker solution concept strengthens what we view as the most important conclusion of [Theorem 1](#), namely that all equilibria attain EFE when [Condition FEI](#) fails.

3.2. Condition FEI and Full Effort

The equivalence between [Condition FEI](#) and the existence of FE equilibria stems from there being no learning in an FE equilibrium. So there is an FE equilibrium if and only if there is one in an auxiliary game without commitment types. It turns out that for a small enough replacement cost, it is further equivalent to consider the auxiliary game without a replacement cost. As mentioned earlier, [Condition FEI](#) characterizes when there are FE equilibria in that auxiliary game; the logic follows standard recursive reasoning ([Abreu, Pearce, and Stacchetti, 1990](#)). In fact, when [Condition FEI](#) fails and replacement costs are sufficiently small, there is a positive lower bound (across all equilibria) on the probability with which opportunistic incumbents shirk in their first period in office; see [Lemma 17](#) in the Appendix. This implies that when [Condition FEI](#) fails, there is some $\varepsilon > 0$ such that no equilibrium has $\mathbb{P}(\mathbf{a}_t = 1) \geq 1 - \varepsilon$ for all $t \in \mathbb{Z}_{\geq 0}$; indeed, the inequality already fails for $t = 0$.

3.3. What Prevents Eventual Full Effort?

Our argument that there is an equilibrium without EFE when [Condition FEI](#) holds is constructive. Notice that when replacement and effort costs are small, and monitoring is

very precise, a newly-installed incumbent has to work with positive probability: if not, a voter would retain an incumbent if he has a favorable reputation (i.e., a posterior π_t strictly above the prior π_0), which would incentivize newly installed incumbents to work toward such a reputation. At least when the replacement cost is zero, a newly-installed incumbent cannot work with certainty either: otherwise, voter incentives imply that any retained incumbent must be working, which implies EFE.

With that in mind, let us describe the non-EFE equilibrium we construct. For simplicity, suppose there are only two signals: a “Pass” signal and a “Fail” one, with the former having the higher likelihood ratio $f_1(s)/f_0(s)$. Opportunistic incumbents work with interior probability in their first period in office. Thereafter, if they have always generated a Fail signal, they are replaced with an interior probability and, if retained, work with an interior probability.¹⁶ At any history with only Pass signals, the incumbent is retained and works. At histories where a Pass signal is followed by a Fail signal, the incumbent is replaced. See [Figure 1](#) for an automaton depiction of the strategy profile. We defer to the [Appendix](#) the explanation of why this profile constitutes an equilibrium. Crucially, however, this equilibrium has the features that every incumbent is eventually replaced, and incumbents in their first period shirk with positive probability. Hence, the equilibrium does not attain EFE; in fact, almost surely there is shirking infinitely often, and average voter welfare (even gross of replacement costs) is below first best:

$$\limsup_{T \rightarrow \infty} \mathbb{E} \left[\frac{1}{T} \sum_{t=0}^{T-1} \mathbf{a}_t \right] < 1.$$

The mechanism underlying the above equilibrium is a novel rationale for undesirable long-run outcomes in accountability models. In particular, it ought to be contrasted with [Myerson \(2006\)](#). For some parameters of his model, his Theorem 1 exhibits equilibria in which opportunistic incumbents always shirk.¹⁷ Yet incumbents are never replaced (and hence EFE does not obtain) because the cost of replacement is large relative to the probability

¹⁶ Since reputation is decreasing in the number of Fail signals, the opportunistic type’s probability of work increases to keep voter incentives fixed. Meanwhile, the voter’s replacement probability is constant to keep politician incentives fixed. Although the underlying forces are quite different, the idea that opportunistic incumbents misbehave less over time is reminiscent of the literature on the repeated prisoner’s dilemma with voluntary separation (e.g., [Ghosh and Ray, 1996](#); [Eeckhout, 2006](#)) and the “starting-small” literature (e.g., [Watson, 1999](#)).

¹⁷ [Myerson’s \(2006\)](#) model for unitary democracy is like ours, except that he assumes a single long-lived and forward-looking voter who can perfectly observe the incumbents’ effort choices, and he allows for large replacement costs. But the relevant part of his Theorem 1 also holds with short-lived voters who can only observe noisy signals, as in our model.

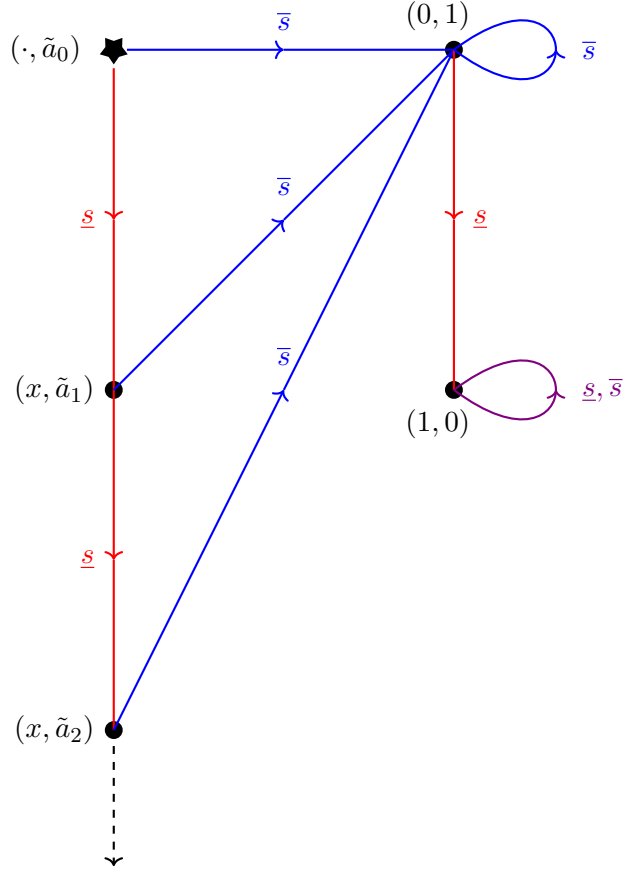


Figure 1: Under [Condition FEI](#), an equilibrium that does not attain eventual full effort when, for simplicity, there are only two signals, $\underline{s}$ and $\bar{s}$, with $f_1(\underline{s})/f_0(\underline{s}) < f_1(\bar{s})/f_0(\bar{s})$. Each node (black star or dot) represents an automaton state, with the initial state—corresponding to a new incumbent—indicated by the star. The vector at each node shows the corresponding strategy profile: the first entry is the probability that the incumbent is replaced (not defined at the initial state), and the second entry is the probability that an opportunistic incumbent exerts effort if retained. Each edge represents the state transition following the corresponding signal. The equilibrium satisfies $0 < \tilde{a}_0 < \tilde{a}_1 < \dots < 1$ and $x \in (0, 1]$.

of a good type. We are interested, instead, in small replacement costs. In our non-EFE equilibrium, voters replace incumbents at some histories despite a favorable reputation (i.e., when the posterior π_t strictly exceeds the prior π_0) because of the rational expectation that an opportunistic incumbent at such a history will shirk.

In fact, this phenomenon of replacing an incumbent despite a favorable reputation is a necessary feature of any equilibrium that does not attain EFE. Formally, we have:

Proposition 2. *Any equilibrium that does not attain eventual full effort entails a positive probability of an incumbent being replaced when his reputation is strictly greater than π_0 .*

An intuition for the argument is as follows. As we will see shortly ([Proposition 3](#), part 1), any opportunistic incumbent is eventually replaced. Hence, if EFE fails, then any good incumbent must also eventually be replaced—otherwise, eventually some good incumbent will stay in office forever, yielding EFE. Moreover, at least absent a replacement cost, there must be some learning in an incumbent’s first period.¹⁸ It follows that incumbents must be replaced at some histories with favorable reputation.

3.4. What Guarantees Eventual Full Effort?

We now turn to why a failure of [Condition FEI](#) implies that all equilibria attain EFE. The key is that precisely because the environment is not conducive to incentive provision, selection is now ineluctable. The following result, a key input to [Theorem 1](#), formalizes the point.

Proposition 3. *Consider any equilibrium.*

1. *Every opportunistic politician who takes office is eventually replaced.*
2. *If [Condition FEI](#) fails, then every good politician who takes office has a positive probability of being retained forever.*

Hence, if [Condition FEI](#) fails, then some good politician is retained forever.

The last statement of [Proposition 3](#) follows from the first two parts and an application of the Borel-Cantelli lemma.

The logic for the first part of the proposition is reminiscent of [Cripps, Mailath, and Samuelson \(2004\)](#). Here is the intuition. Since any incumbent’s reputation is a martingale, it converges almost surely. If an opportunistic incumbent is not eventually replaced, then

¹⁸ Otherwise, a new incumbent must be working, which implies FE when there is no replacement cost.

because signals statistically identify effort, eventually either (i) he must be exerting effort—behaving like the good type, so that belief updating stops—or (ii) his type must be revealed. Case (i) contradicts him not being eventually replaced, because it is only the possibility of replacement that provides effort incentives. The same logic applies if we are in case (ii) and yet the incumbent eventually still exerts effort with positive probability. Finally, if we are in case (ii) and the incumbent is certain to shirk, then it is optimal for voters to replace him.

Now consider the second part of [Proposition 3](#). When [Condition FEI](#) fails, full-effort incentives cannot be sustained; in particular, as noted earlier, a newly installed incumbent shirks with positive probability. Thus, once an incumbent’s reputation is sufficiently high, a voter strictly prefers retention to replacement, since replacement exposes her to some chance of shirking. Put differently, there is a reputation threshold above which the voter never replaces. The key step is that these high no-replacement beliefs are reached with positive probability. In fact, we show that reputation can get arbitrarily high. The logic is that if the supremum of on-path beliefs were interior, then at beliefs close to that supremum an opportunistic incumbent would have to exert effort with high probability—otherwise some signal would push the posterior above the supposed supremum, contradicting that being a supremum. But such effort is intuitively at odds with the failure of [Condition FEI](#),¹⁹ and so the supremum must equal one. Now, once beliefs enter the high region where there is no replacement, martingale arguments imply that there is a positive probability of never leaving that region. Combined with the first part of the proposition, it follows that every good-type officeholder is retained forever with positive probability when [Condition FEI](#) fails.

[Proposition 3](#) implies that when [Condition FEI](#) fails, every equilibrium asymptotically yields voters their first best welfare, even net of replacement cost. More precisely:

Corollary 1. *Suppose [Condition FEI](#) fails. In any equilibrium, the period- t voter’s expected payoff converges to first best as $t \rightarrow \infty$.*

Although [Proposition 3](#) shows that selection is extremely powerful in the long run in every equilibrium (when [Condition FEI](#) fails), it prompts a question about the short run. To assure that opportunists are eventually replaced, what makes replacement attractive to voters in the first place—especially when good types may be rare? One might guess that the voters’ endogenous **outside option**—their equilibrium payoff from replacing an incumbent, gross of replacement cost—must be high for selection to operate effectively.²⁰ In fact, the following

¹⁹ This is only an intuition because [Condition FEI](#) does not actually preclude the possibility of effort with certainty at some histories. The formal argument is substantially more involved.

²⁰ More precisely, we define the outside option as the probability of effort from a newly-installed incumbent, which can be written as $\mathbb{E}[\mathbf{a}_0] = \alpha(\emptyset) = \pi_0 + (1 - \pi_0)\sigma_P(\emptyset)$.

result says that the exact opposite is true when good types are scarce and replacement costs are small.

Proposition 4. *Suppose [Condition FEI](#) fails. For any $\varepsilon > 0$, there exist $\bar{c} > 0$ and $\bar{\pi}_0 \in (0, 1)$ such that, if $c \leq \bar{c}$ and $\pi_0 \leq \bar{\pi}_0$, then every equilibrium has voters' outside option below ε .²¹*

In other words, assuming [Condition FEI](#) fails, the combination of scarce good types and low replacement costs implies that voters' outside option is arbitrarily low—or equivalently, newly-installed opportunists are almost certainly shirking. So, replacement can look arbitrarily unattractive in the short run, despite culminating powerfully in the long run. While we defer the logic of [Proposition 4](#) to the [Appendix](#), why does a low outside option not undermine eventual selection? It is because opportunists are assured to reach histories at which voters believe the incumbent is overwhelmingly likely to be an opportunist who will shirk—so much so that even the low outside option is attractive enough to trigger replacement, given small replacement costs.

4. Discussion

This section discusses some extensions of our model.

Bad types. The possibility that some politicians are good types—intrinsically motivated to exert effort in office—is essential to our results. If all politicians are opportunistic, then although [Condition FEI](#) still characterizes when full effort can be attained, there is always an equilibrium (regardless of parameters) in which incumbents always shirk.

What if we allow for the coexistence of good types with *bad* types who always shirk in office? In addition to suitably adapting our maintained assumption of replacement costs being small enough, [Condition FEI](#) must also be strengthened for the existence of an equilibrium in which opportunistic incumbents always exert effort. The reason is that now some sequences of signals will convince voters that the incumbent is likely a bad type and must be replaced, which makes it more difficult to sustain effort from the opportunistic type along such sequences. More interestingly however, when [Condition FEI](#) fails it remains true that all equilibria attain eventual full effort. The mechanism still operates via selection of the good type, with an argument analogous to that of [Proposition 3](#). In particular, following the logic of [Cripps, Mailath, and Samuelson \(2004\)](#), not only will every opportunistic officeholder eventually be replaced, but so too every bad type.

²¹ Recall that [Condition FEI](#) is a condition only on (κ, δ, f) , not on (c, π_0) . We show that for any (κ, δ, f) violating [Condition FEI](#), an appropriate pair $(\bar{c}, \bar{\pi}_0)$ can be found to deliver the conclusion.

Exogenous turnover. We have assumed that politicians remain eligible for office indefinitely. Plainly, this is essential for [Proposition 3](#)’s conclusion that when [Condition FEI](#) fails, some good type will eventually stay in office forever.

Suppose now that incumbents exit office exogenously with probability $1 - \rho$ in each period (e.g., due to retirement or death), for some survival probability $\rho \in (0, 1)$. Beyond implying that every incumbent leaves office at some time almost surely, politicians’ discount factors are also effectively reduced to $\delta\rho$. Naturally, then, the relevant version of [Condition FEI](#) now replaces δ with $\delta\rho$ in (IC_{FEI}) and (PK_{FEI}) . Our full-effort equilibrium construction remains valid when this version of [Condition FEI](#) holds. However, when this version of [Condition FEI](#) fails, eventual full effort cannot be attained in any equilibrium—the eventual exit of every incumbent creates a friction even after adjusting for the lower effective discount factor. Nevertheless, our conclusion regarding the power of selection is robust in the following sense. Holding the effective discount factor $\delta\rho$ fixed—so, in particular, not altering the aforementioned version of [Condition FEI](#)—we can ask what happens as $\rho \rightarrow 1$. Because voters are short-lived and politician incentives depend only on the effective discount factor, the set of equilibrium strategies does not change. Thus, the equilibrium distributions of the “search time” for a good type to take office and never be replaced (barring exogenous exit) is unchanging. As the survival probability $\rho \rightarrow 1$, in every equilibrium the system spends a fraction of time approaching 1 in the hands of a good type, and the long-run average effort probability converges to 1.

Recalling incumbents. We have assumed that an incumbent who is replaced can never hold office again. This “no-recall” assumption is commonplace in the literature, dating back to [Banks and Sundaram \(1993\)](#); it was also discussed by [Ferejohn \(1986\)](#) in his pure moral-hazard setting. The assumption substantially simplifies tracking the voters’ outside option by making it history independent (given the focus on [personal equilibria](#)). Even if recall is allowed, [Condition FEI](#) still characterizes when there is a full-effort equilibrium, and when the condition holds, our construction of an equilibrium without eventual full effort goes through as well. While one might conjecture that allowing for recall would also not change our result that all equilibria attain eventual full effort when [Condition FEI](#) fails, we do not have a proof.

A related point is that we assume an infinite pool of politicians. With a finite pool, one would have to consider recall. Moreover, for selection to be assured in the long run, one would have to assume the finite pool contains at least one good type—so politician types

could not be independent. Such an analysis is beyond the scope of the current paper.²²

Long-lived voter. As discussed in [Section 2](#), there are multiple reasons for our assumption of short-lived voters. But now consider instead only one long-lived voter who maximizes her discounted average payoff with a discount factor $\delta_V \in [0, 1)$, which can be different from the politicians’ discount factor δ .

When (δ, κ, f) satisfies [Condition FEI](#) and the replacement cost c is small, there is still an equilibrium that attains full effort as well as one that fails eventual full effort. The former requires no change from our full-effort equilibrium construction. However, in our equilibrium that fails eventual full effort ([Figure 1](#)), a politician’s strategy now needs to be modified to account for the voter’s discount factor δ_V . Specifically, at non-initial career histories consisting only of Fail signals, the probabilities with which the opportunistic type works in order to keep the voter indifferent depend on δ_V ; but suitable probabilities exist.

When [Condition FEI](#) is violated, there is still no equilibrium with full effort. However, it is unclear whether eventual full effort attains in all, or even some, equilibria. In particular, it is unclear whether equilibria have a positive probability of retaining each good-type officeholder in the long run, because we cannot rule out that equilibrium reputation is bounded away from 1.²³ This echoes the difficulty of obtaining tight predictions on equilibrium payoffs in reputation models without replacement but with two long-lived players ([Cripps and Thomas, 1997](#); [Chan, 2000](#)).

²²In ongoing work, [Ali \(2025\)](#) studies a model similar to ours, except that he has only two politicians. He identifies conditions under which every equilibrium within a class attains eventual full effort, and shows some of those equilibria attain full effort. By contrast, in our setting [Theorem 1](#) says these two desiderata hold only under disjoint conditions. [Ali’s \(2025\)](#) eventual full effort logic is not one of selection, unlike ours. One reason for the difference is [Ali’s \(2025\)](#) focus on pure Markov strategies for incumbents. Our equilibrium that fails eventual full effort entails incumbents mixing; indeed, [Proposition 2](#) implies that every equilibrium of our model in which politicians use pure strategies yields eventual full effort. We see no compelling reason to exclude mixed strategies from consideration; inter alia, as in many models of reputation, one cannot otherwise assure equilibrium existence for all parameters—specifically, when our [Condition FEI](#) fails.

²³In more detail: with a long-lived voter, it is no longer the case that the incumbent needs to be retained with positive probability following some signal s with $f_1(s) > f_0(s)$. As an example, consider a career history h at which the incumbent shirks: $\sigma_P(h) = 0$. Unlike when the voter is short-lived, the incumbent being retained with positive probability at h no longer implies that he will be retained with probability 1 at any career history h' with higher reputation ($\pi(h') > \pi(h)$), as a long-lived voter’s discounted average payoff from keeping this incumbent can be lower at h' even when the voter’s stage-payoff from doing so is higher at h' .

5. Conclusion

Our motivating question is whether the power of replacement ensures that officeholders act in voters' interests in the long run. Inspired by previous models of accountability and reputation, we have studied the question in a simple and stylized model, one that we believe is a natural benchmark. Our analysis yields a sharp, dichotomous answer: when there are some intrinsically-motivated politicians, replacement makes good long-run outcomes always possible, but only guaranteed in environments that are *not* conducive to incentive provision.

Our main result, [Theorem 1](#), provides a new perspective on the tension between sanctions and selection that has been noted by others (e.g., [Banks and Sundaram, 1993](#); [Fearon, 1999](#)). In environments conducive to incentive provision, there is an equilibrium with perpetual effort from incumbents. But politics can alternatively fall into a trap of excessive turnover, driven by voters' rational replacement of incumbents whose favorable past performance now makes them shirk, which bounds long-run outcomes away from full effort. In these environments, it is not the underlying conditions that determine long-run outcomes; instead political norms—self-fulfilling expectations—do, even absent any intertemporal voter coordination.

Conversely, in environments where strong incentives cannot be provided (i.e., there is no equilibrium with full effort), the very weakness of accountability becomes the engine of its long-run success. Selection is now inexorable: opportunistic types are weeded out, and voters are assured to eventually secure a good politician who works in their interest. To return to the view of J.S. Mill quoted in the introduction, in our model it is precisely when opportunistic officeholders cannot be fully disciplined that the worth of the State is eventually determined by the worth of the individuals composing it.

A. Proof Appendix

In our proofs below, we do *not* maintain the assumption made in the main text that $c < \min\{\pi_0, 1 - \pi_0\}$. Instead, to clarify which results require what (if any) assumption on the replacement cost c , we state the requisite assumption when needed.

A.1. Equilibrium Existence

As discussed in [Section 3.1](#), our solution concept can be strengthened to a **personal symmetric perfect Bayesian equilibrium (PBE)** by dropping the qualification of $\sigma_V(h) < 1$ in condition (3) of our [equilibrium definition](#). The proof below shows that such a PBE exists in our game.

Proof of Proposition 1. Let $\mathcal{H}$ be as defined for our original game. For any $\varepsilon \in [0, \frac{1}{2}]$ and $T \in \mathbb{N} \cup \{\infty\}$, define an auxiliary game $\Gamma(\varepsilon, T)$ with a voter in every period and a single politician—who will be a good type with probability π_0 and an opportunistic type with probability $1 - \pi_0$.

We begin by describing $\Gamma(0, \infty)$. The game form is almost exactly as in our original game before the first time a voter replaces the politician, except that at the initial history $h = \emptyset$, the politician chooses a continuous action $\tilde{a}_0 \in [0, 1]$, which generates a signal from $f_{\tilde{a}_0} \in \Delta S$. Say that a politician *semi-pure* strategy is $\sigma_P : \mathcal{H} \rightarrow [0, 1]$, with the interpretation that the politician is not mixing at the initial history but could be mixing (over the binary action set $\{0, 1\}$) at other histories. Voters' (mixed) strategies are given by $\sigma_V : \mathcal{H} \times S \rightarrow [0, 1]$. Furthermore, in this modified game, replacing the politician ends the game. The payoffs of the politician are exactly the same as those of the initial officeholder in our original model. If the politician is not replaced, the payoffs of voters in our auxiliary game are exactly the same as those in our original model. But if a voter replaces the politician, then her payoff is equal to $\pi_0 + (1 - \pi_0)\tilde{a}_0$, a function of the unobserved initial choice of the politician.

A key feature of the game $\Gamma(0, \infty)$ is that the politician's choice $\tilde{a}_0 \in [0, 1]$ directly enters the payoff of the voter who replaces the politician. That is why we treat the politician's choice at the initial history as a continuous action rather than a randomization over the actions 0 and 1.²⁴ With that caveat, we note that every PBE of the auxiliary game $\Gamma(0, \infty)$ in which the politician uses a semi-pure strategy—a semi-pure PBE, for short—corresponds

²⁴If the politician were to mix between actions 0 and 1 (or any other actions) at the initial history, subsequent voters' incentives would depend on the signal s_0 , because it would convey information about the realized initial action, and so the value of replacing the politician.

exactly to a personal symmetric PBE of our original model.²⁵

So it remains to see that $\Gamma(0, \infty)$ has a semi-pure PBE. The remainder of the proof now follows familiar lines (à la [Fudenberg and Levine \(1983\)](#) and [Kreps and Wilson \(1982\)](#)), but we summarize such an argument for the sake of completeness because the game $\Gamma(0, \infty)$ is not an instance of the models covered in those papers.

To show existence, define for every time horizon $T \in \mathbb{Z}_{>0}$ the game $\Gamma(\varepsilon, T)$ by further constraining players' behavior as follows. Before time T , all players (specifically, all voters and opportunistic-type politicians) choose from $[\varepsilon, 1 - \varepsilon]$ at every history rather than from $[0, 1]$; and starting at time T , all players choose from $\{\frac{1}{2}\}$.

First, consider the case of $\varepsilon > 0$ and $T < \infty$. Let us view the game $\Gamma(\varepsilon, T)$ in agent form (with a politician agent and voter agent at each public history). Observe that each agent chooses from a compact interval, only finitely many agents make a nontrivial choice, payoffs are continuous in the strategy profile, and each agent's payoff is affine in their own action. Therefore, by the Debreu-Fan-Glicksberg theorem, a Nash equilibrium $(\sigma_P^{\varepsilon, T}, \sigma_V^{\varepsilon, T})$ exists for $\Gamma(\varepsilon, T)$. Because every history is reached with strictly positive probability under this profile, a unique belief map $\pi^{\varepsilon, T}$ exists that satisfies the Bayesian property given $\sigma_P^{\varepsilon, T}$. Also because every history is reached with strictly positive probability, the Nash equilibrium property implies sequential optimality (in the restricted game).

Now, fixing $\varepsilon > 0$, consider the sequence $(\sigma_P^{\varepsilon, T}, \sigma_V^{\varepsilon, T}, \pi^{\varepsilon, T})_{T=1}^{\infty}$ from the compact metrizable space $\left[([\varepsilon, 1 - \varepsilon] \times [\varepsilon, 1 - \varepsilon]^S \times [0, 1])^{\mathcal{H}}\right]^3$. By compactness, some subsequence converges to some triple $(\sigma_P^{\varepsilon}, \sigma_V^{\varepsilon}, \pi^{\varepsilon})$.

Appealing again to compactness, $(\sigma_P^{\varepsilon}, \sigma_V^{\varepsilon}, \pi^{\varepsilon})_{\varepsilon \in (0, \frac{1}{2})}$ also has some limit point $(\sigma_P, \sigma_V, \pi)$ as $\varepsilon \rightarrow 0$. Continuity of payoffs in the strategy profile (given that the game is continuous at infinity) and continuity of the Bayesian condition (which is easy to see by clearing the denominator) imply $(\sigma_P, \sigma_V, \pi)$ is a semi-pure PBE of $\Gamma(0, \infty)$. $\square$

A.2. The Main Theorem

A.2.1. Proof Overview

As the main theorem's proof is somewhat involved, we first provide an overview of the arguments. To help parse the logical flow of these arguments, [Figure 2](#) details which results

²⁵ To be precise, our notion of PBE in the auxiliary game is what [Mailath \(2018\)](#) refers to as an “almost perfect Bayesian equilibrium”—although [Mailath \(2018\)](#) restricts attention to finite games, which our auxiliary game is not, the definition can be applied essentially verbatim because we have a finite set of signals.

are inputs to which other results in the paper.

Preliminary calculations. [Appendix A.2.2](#) begins with some preliminary calculations. Adapting standard repeated-games arguments, [Lemma 1](#) confirms that [Condition FEI](#) exactly captures the ability to incentivize politicians to exert effort whenever in office. Using a compactness argument, [Lemma 2](#) notes that, if [Condition FEI](#) fails, it fails uniformly. More specifically, the lemma finds a lower bound on how much a politician’s continuation value must be allowed to grow (following some signal realizations) in order to make working incentive compatible. Applying this bound, [Lemma 3](#) finds a finite sequence of signals during which the incumbent must strictly prefer to shirk at least once.

Eventual full effort. In [Appendix A.2.3](#), we establish that a failure of [Condition FEI](#) generates eventual full effort when replacement costs are low. The key mechanism is one of selection ([Proposition 3](#)): opportunistic types are always replaced, whereas good types have a positive probability of being kept forever, meaning the latter is sure to happen eventually. (In [Appendix A.3](#), we observe that this selection easily implies a welfare statement—[Corollary 1](#), which says the time- t voter’s payoff converges to first best as $t \rightarrow \infty$.)

Let us detail the argument that opportunistic types are eventually replaced. First, [Lemma 4](#) yields a finite time horizon and a lower bound on the incumbent’s probability of being replaced over that horizon starting from any history at which an opportunistic incumbent exerts effort. Indeed, permanent shirking being unprofitable imposes a lower bound on the probability of being replaced over a long enough finite horizon. But then full-support monitoring lets us convert this lower bound on the firing rate given this deviation to a lower bound on the on-path firing rate. Then, [Lemma 5](#) says that a given officeholder is eventually replaced, has their type asymptotically revealed, or asymptotically behaves indistinguishably from the good type. The reasoning is as in [Cripps, Mailath, and Samuelson \(2004\)](#). By the martingale convergence theorem, reputation converges (almost surely). Given that monitoring statistically identifies the effort choice, beliefs can have an interior limit with infinitely many signals only if effort is asymptotically uninformative of the officeholder’s type—that is, if the opportunistic type behaves the same way as the good type in the limit.

Next, [Lemma 6](#) shows that, in fact, a given officeholder is eventually replaced or has their type asymptotically revealed. In light of [Lemma 5](#), the only case to rule out is that in which that officeholder is never replaced and, yet, asymptotically exerts effort with probability approaching 1. But in the latter event, it is eventually optimal for the officeholder to work in every period, and so [Lemma 4](#) ensures (by bounding the average hazard rate of firing below) that the officeholder is eventually replaced. Next, if the cost of replacement is low, [Lemma 7](#)

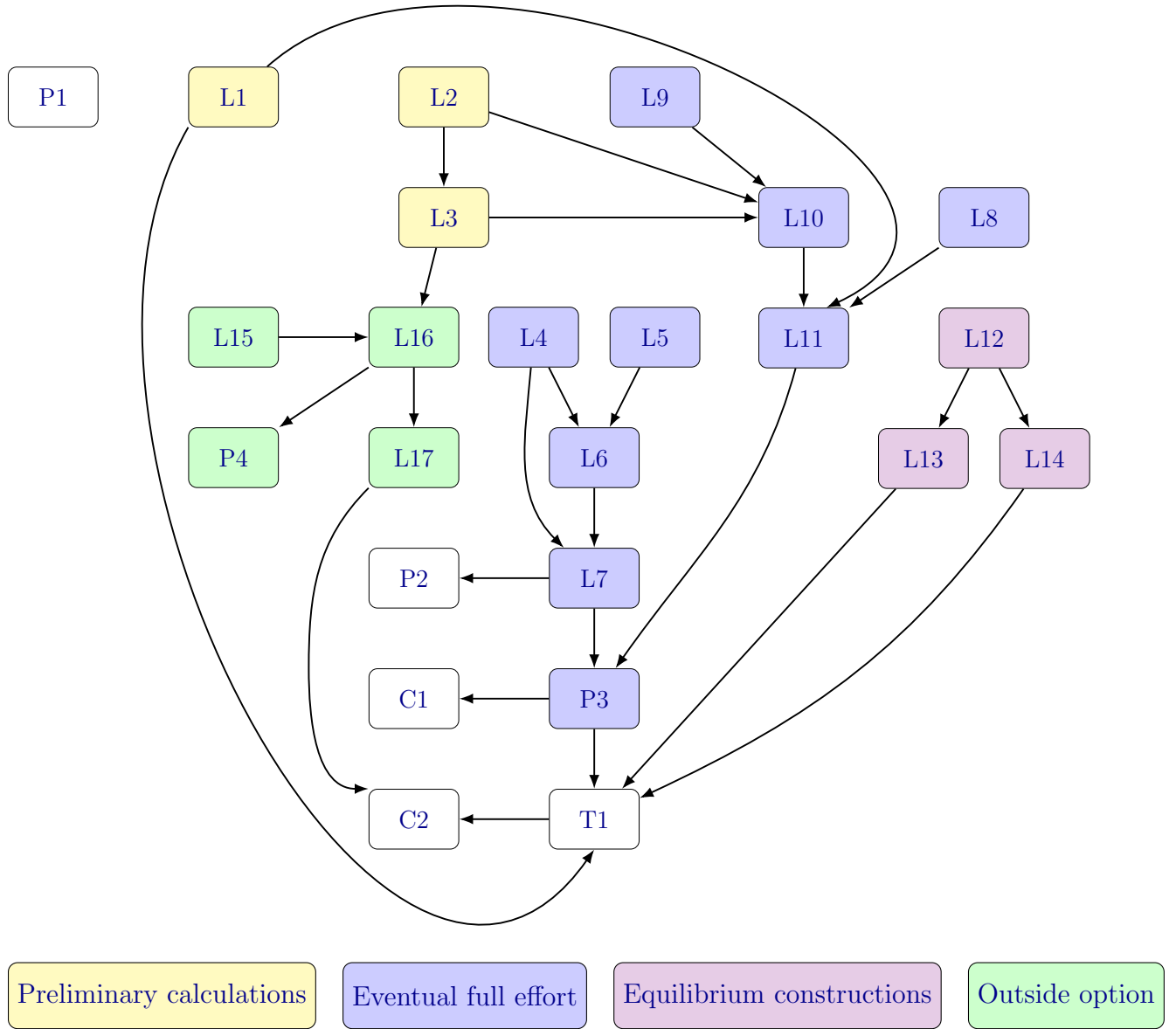


Figure 2: This flow chart depicts the logical relationships between every lemma (L), theorem (T), corollary (C), and proposition (P). An arrow pointing from one result to another indicates that the former is a direct input to the latter. Different colors correspond to different sections of the proof appendix.

shows that opportunistic types are always eventually replaced. Because of [Lemma 4](#), the only thing to prove is that an officeholder cannot be indefinitely retained with voters being asymptotically certain he is opportunistic. We rule this possibility in two exhaustive cases. First, if the opportunistic type is eventually expected to exert effort with positive probability, then [Lemma 4](#) again guarantees eventual replacement. Second, in the complementary case, the officeholder at some point has a very low reputation and is expected to shirk; but in this case, the voter would strictly prefer to replace and face a fresh draw.

We next argue that, if [Condition FEI](#) fails, good types have a positive probability of never being replaced. To show it, [Lemma 8](#) establishes that a high enough reputation on the path of play—specifically, one such that the current voter would want to keep the incumbent even if the opportunistic type were expected to shirk—leads to a positive probability of being retained forever. The argument follows two steps. First, because beliefs are a martingale, Doob’s optional stopping theorem tells us that once above this threshold they have a positive probability of permanently staying above this threshold (hence, by voter incentives, the incumbent is never replaced). Second, a positive ex-ante probability of this permanent-retention event implies a positive joint probability of both being retained forever and being a good type. Now, [Lemma 9](#) says that any on-path career history (that is, one with a positive probability of an incumbent reaching it without being replaced) is followed by some immediately subsequent on-path history with an additional “good” signal (that is, a signal strictly indicative of work). The lemma is proven by considering two cases. First, if the incumbent shirks with positive probability at the current history, then there is learning: following any good signal, the next period’s voter will have a strictly stronger incentive than this period’s voter to keep the incumbent, and so will do so in equilibrium. Second, if the incumbent works at the current history, his incentive to do so implies he will be rewarded (hence have strictly positive continuation value) for some good signal.

[Lemma 10](#) then says an incumbent’s reputation can become arbitrarily good at some on-path histories if [Condition FEI](#) fails. The key to establishing this fact is to consider play at on-path histories with beliefs very close to a supremum that is putatively interior. As [Lemma 9](#) says some good signal results in a positive probability of retention, the hypothesis that we are near an interior supremum means opportunistic types work with high probability. But then, there is little learning no matter the signal, so next period’s belief is again close to that supremum. Iterating this argument over the (finite) horizon identified by [Lemma 3](#) generates a history at which reputation is close to the interior supremum and yet the incumbent shirks, a contradiction.

[Lemma 11](#) finally argues that any good officeholder has a positive probability of perpetual

retention when [Condition FEI](#) fails. In light of [Lemma 10](#), the result is immediate as long as the threshold required in [Lemma 8](#) is strictly below 1. It obviously is when the replacement cost is strictly positive. With a 0 replacement cost, we just need to see that the voters' outside option is strictly less than 1 in any equilibrium. But an equilibrium with outside option 1 would have incumbents working at every on-path history, which [Lemma 1](#) tells us cannot happen when [Condition FEI](#) fails.

Equilibrium constructions. [Appendix A.2.4](#) constructs equilibria with different properties when [Condition FEI](#) holds. As observed in [Lemma 1](#), [Condition FEI](#) is equivalent to the existence of a voter strategy (ignoring voter incentives) that makes incumbents willing to work in every period until they are replaced. [Lemma 12](#) says that if such a voter strategy exists, then there is one with a special form: a time-and-history-independent likelihood ratio test, in which every “good” signal (one that is indicative of work), and possibly some “bad” signals, pass the test. To see why, we start with such a voter strategy that incentivizes full effort (which without loss gives its highest possible continuation value to a new officeholder) and successively modify it until it takes the desired form. We first argue that every good signal results in the highest continuation value in any period. The reason is that rewarding a successful incumbent both relaxes contemporaneous incentive constraints and makes the relationship more valuable—the latter relaxing past incentive constraints. Second, a simple replication argument shows that treating all signals with the same likelihood ratio f_1/f_0 equally preserves incentives. Third, making the continuation value more assortative with the likelihood ratio, while preserving the on-path value of the officeholder, only amounts to higher-powered incentives. The voter strategy thus takes the form of a likelihood-ratio test: replace the incumbent for producing a likelihood ratio below a cutoff, keep her with fixed high probability for a likelihood ratio above the cutoff, and possibly use an intermediate probability if the likelihood ratio is equal to the cutoff. Next, because the incentive constraint is weakly relaxed *either* by lowering the intermediate probability or by raising the intermediate probability, it follows that the mixing is redundant. Finally, because lowering the firing rate for “passing” the likelihood ratio test (which amounts to scaling up continuation values) relaxes all incentive constraints, we can further assume the incumbent is replaced only for producing a bad signal. With this lemma in hand, we can explicitly construct various equilibria.

[Lemma 13](#) constructs an equilibrium that attains full effort. On path, the officeholder always works, and the voter uses the strategy detailed by [Lemma 12](#). If any voter was supposed to have replaced a politician but did not, then this politician always shirks thereafter,

her reputation is permanently zero, and all future voters replace him.²⁶ As long as the replacement cost is low, voters find it incentive compatible to replace the incumbent when they should, and every other incentive constraint holds by construction.

The last required input to the theorem is [Lemma 14](#), which constructs an equilibrium that does not attain eventual full effort. Again, the equilibrium builds on the likelihood ratio test of [Lemma 12](#). Call a signal a Pass if it passes the ratio test and a Fail otherwise; all Fail signals are bad, in the sense of being strictly indicative of shirking, as explained above in the discussion of [Lemma 12](#). For a given small enough replacement cost, the construction—illustrated in [Figure 1](#) for the case with binary signals—is as follows. Right after the first time a Pass signal is produced, the continuation profile looks exactly like the construction of [Lemma 13](#). In particular, on path, the opportunistic type of officeholder works hard until she is replaced, and she is replaced the next time she produces a Fail signal. Before this time (i.e., when only bad signals have been produced so far), voters replace the officeholder with a fixed probability that is chosen to make the politician indifferent between working and shirking; this probability exists by ([Lemma 12](#) and) the definition of Pass and Fail signals. The opportunistic type’s initial probability of effort is strictly below 1, but is chosen to be high enough that for the relevant range of reputations, a voter who expects an opportunistic incumbent to shirk is willing to bear the replacement cost. In the rest of the first regime, before the first Pass signal, we recursively describe the officeholder’s effort by making the voter indifferent between keeping and replacing at every such history. This is possible to do because (due to Fail signals being bad) his reputation is falling, which can be offset exactly by some increase in the effort probability from opportunistic types. Finally, we verify that the construction is an equilibrium, and observe that it does not attain eventual full effort because every incumbent is eventually replaced (as he will at some point produce a Pass signal followed by a Fail signal) and so voters never stop being exposed to the initial probability of shirking.²⁷

A.2.2. Preliminary Calculations

[Table 2](#) summarizes some of the key notation used in the formal arguments that follow.

We begin by confirming that [Condition FEI](#) indeed captures incumbents’ incentives to exert full effort. Although this fact follows directly from adapting standard self-generation arguments ([Abreu, Pearce, and Stacchetti, 1990](#)), we include a proof for completeness.

²⁶ When the replacement cost is positive ($c > 0$), the belief at the first off-path history in this construction is valid because we are using *weak* PBE. When $c = 0$, our construction can be readily modified to be a PBE.

²⁷ This further implies that, as mentioned in [Section 3.3](#), average voter welfare is strictly below first best even gross of the replacement cost.

Lemma 1. *The following are equivalent:*

1. *Some strategy profile attains full effort and satisfies politician incentive compatibility.*
2. *Some equilibrium of the modified game with $(\pi_0, c) = (0, 0)$ attains full effort.*
3. *Condition FEI holds.*

Proof. Let us first show condition 1 implies condition 3. To that end, suppose the strategy profile (σ_P, σ_V) attains full effort and satisfies politician incentive compatibility. Let $v : \mathcal{H} \rightarrow [0, 1]$ capture the politician's continuation values under this strategy profile, and let $\mathcal{H}^* \subseteq \mathcal{H}$ denote the set of on-path histories. Consider the set

$$V := \{ (v(h), ([1 - \sigma_V(h, s)] v(h, s))_{s \in S}) : h \in \mathcal{H}^* \} \subseteq [0, 1 - \kappa] \times [0, 1 - \kappa]^S,$$

and let $V_0 \subseteq [0, 1]$ denote the projection of V onto its first coordinate. These sets are nonempty because $\emptyset \in \mathcal{H}^*$. So let $\bar{v}_0 := \sup V_0 \in [0, 1]$. By definition, V contains some sequence $(v_0^n, (v^n(s))_{s \in S})_{n=1}^\infty$ such that $\lim_{n \rightarrow \infty} v_0^n = \bar{v}_0$. Dropping to a subsequence, we may assume (because $[0, 1]$ is compact) that $v^n = (v^n(s))_{s \in S}$ also converges to some $\bar{v} \in [0, 1]^S$ as $n \rightarrow \infty$. Let us show that $\bar{v}$ witnesses Condition FEI. Because it is incentive compatible for the politician to work at every history in $\mathcal{H}$, every n has $v_0^n = (1 - \delta)(1 - \kappa) + \delta f_1 \cdot v^n \geq (1 - \delta) + \delta f_0 \cdot v^n$, so that taking limits yields

$$\bar{v}_0 = (1 - \delta)(1 - \kappa) + \delta f_1 \cdot \bar{v} \geq (1 - \delta) + \delta f_0 \cdot \bar{v}.$$

Meanwhile, any $h \in \mathcal{H}^*$ and $s \in S$ have either $\sigma_V(h, s) = 1$ or $(h, s) \in \mathcal{H}^*$ by definition, hence $[1 - \sigma_V(h, s)] v(h, s)$ is a convex combination of 0 and something in V_0 . Thus, every $n \in \mathbb{Z}_{>0}$ and $s \in S$ have $v^n(s) \leq \bar{v}_0$, so that taking limits yields $\bar{v} \in [0, \bar{v}_0]^S$. Hence, $\bar{v}_0 = f_1 \cdot \bar{v} \geq f_0 \cdot \bar{v}$ and $\bar{v}_0 \geq \max \bar{v}(S)$, witnessing Condition FEI.

To see that condition 3 implies condition 2, suppose Condition FEI holds, witnessed by $v : S \rightarrow \mathbb{R}_{\geq 0}$. Letting $v_0 := (1 - \delta)(1 - \kappa) + \delta \sum_{s \in S} f_1(s) v(s) \geq \max_{s \in S} v(s)$, we can construct a full-effort equilibrium as follows. At every history h , the politician chooses $\sigma_P(h) = 1$, the voter's belief is $\pi(h) = 0$, and following each signal $s \in S$ the voter chooses $\sigma_V(h, s) = 1 - \frac{v(s)}{v_0} \in [0, 1]$. The Bayesian property, and the incentives of the voter (who has payoff 1 for either re-election or replacement) are trivial. Toward politician incentives, note that the constant function $h \mapsto v(h) = v_0$ satisfies the recursive equation for politician continuation payoffs. Because a unique such bounded function $\mathcal{H} \rightarrow \mathbb{R}$ exists (by the Banach fixed point theorem), it must be that the politician's continuation value is v_0 at every history. But then, (IC_{FEI}) tells us the politician finds it optimal to work at every history, confirming

equilibrium.

Finally, condition 2 obviously implies condition 1, delivering the lemma. $\square$

The following lemma says that if [Condition FEI](#) fails, it fails in a uniform way.

Lemma 2. *If (κ, δ, f) do not satisfy [Condition FEI](#), then some $T \in \mathbb{Z}_{>0}$ exists such that every $v \in \mathbb{R}_{\geq 0}^S$ satisfying [\(IC_{FEI}\)](#) has $(1 - \delta)(1 - \kappa) + \delta \sum_{s \in S} f_1(s)v(s) < \max_{s \in S} v(s) - \frac{1}{T}$.*

Proof. Suppose [Condition FEI](#) fails.

The set V of $v \in [0, 1]^S$ satisfying [\(IC_{FEI}\)](#) is compact, and the function $v \mapsto \max v(S) - [(1 - \delta)(1 - \kappa) + \delta f_1 \cdot v]$ is continuous on it. Failure of [Condition FEI](#) then implies this function is strictly positive on V , and so bounded below by a strictly positive number.

Meanwhile, any $v \in \mathbb{R}_{\geq 0}^S$ with $\bar{v} := \max v(S) \geq 1$ has

$$(1 - \delta)(1 - \kappa) + \delta f_1 \cdot v \leq (1 - \delta)(\bar{v} - \kappa) + \delta \bar{v} = \bar{v} - (1 - \delta)\kappa,$$

so that

$$\inf_{v \in \mathbb{R}_{\geq 0}^S: v \text{ satisfies } \text{(IC}_{\text{FEI}}\text{)}} \{ \max v(S) - [(1 - \delta)(1 - \kappa) + \delta f_1 \cdot v] \} > 0.$$

Thus $\frac{1}{T}$ is strictly below this infimum value for T large enough. $\square$

The next lemma finds, when [Condition FEI](#) fails, a sequence of signals and a duration over which an incumbent cannot be incentivized to work.

Lemma 3. *Suppose [Condition FEI](#) fails, let T be as delivered by [Lemma 2](#), and fix any equilibrium and any $h_0 \in \mathcal{H}$ with $v(h_0) > 0$. Recursively define $s_0, \dots, s_{T-1} \in S$ and $h_1, \dots, h_T \in \mathcal{H}$ by letting $s_t \in \arg \max_{s \in S} \{[1 - \sigma_V(h_t, s)]v(h_t, s)\}$ and $h_{t+1} = (h_t, s_t)$ for $t \in \{0, \dots, T-1\}$. Then some $t \in \{0, \dots, T-1\}$ has $\sigma_P(h_t) = 0$.*

Proof. Assume for a contradiction that every $t \in \{0, \dots, T-1\}$ has $\sigma_P(h_t) > 0$. For each such t , the fact that the incumbent willingly exert effort tells us the vector $v^t := ([1 - \sigma_V(h_t, s)]v(h_t, s))_{s \in S} \in [0, 1]^S$ satisfies [\(IC_{FEI}\)](#). Hence, [Lemma 2](#) says

$$v(h_t) = (1 - \delta)(1 - \kappa) + \delta \sum_{s \in S} f_1(s)v^t(s) < v^t(s_t) - \frac{1}{T} \leq v(h_{t+1}) - \frac{1}{T}.$$

Therefore,

$$0 \leq v(h_0) < v(h_T) - T \frac{1}{T} \leq 1 - T \frac{1}{T} = 0,$$

a contradiction. $\square$

For the analysis that follows, we introduce some notation to describe voters' belief updating.

Notation 1. For any belief $\pi \in (0, 1]$, work probability $a \in [0, 1]$ for the opportunistic type, and signal $s \in S$, let

$$\beta_a(\pi|s) := \frac{\pi f_1(s)}{[\pi + (1 - \pi)a] f_1(s) + (1 - \pi)(1 - a)f_0(s)} \in (0, 1]$$

denote the updated Bayesian belief that a voter would have after observing signal s .

For any $\pi \in (0, 1]$ and bound $\eta \in [0, 1]$, let

$$\bar{\beta}_\eta(\pi) := \sup_{a \in [\eta, 1], s \in S} \beta_a(\pi|s) \in (0, 1]$$

denote the highest level beliefs can grow to in a period when the opportunistic-type incumbent's expected effort is at least η .

For any $t \in \mathbb{Z}$ and $\eta \in [0, 1]$, let $\bar{\beta}_\eta^t : (0, 1] \rightarrow (0, 1]$ denote the t -fold composition of $\bar{\beta}_\eta$.

A.2.3. Eventual Full Effort

Primarily for the argument that a failure of [Condition FEI](#) implies all equilibria attain eventual full effort, we will also use the following notation.

Notation 2. Given any equilibrium, we define the following random variables.

For any $t \in \mathbb{Z}_{\geq 0}$, let $\mathbf{r}_t \in \{0, 1\}$ be the random variable indicating whether the first officeholder has been replaced as of the beginning of period t ; and let $\tau_{\mathbf{r}} := \inf\{t \in \mathbb{Z}_{\geq 0} : \mathbf{r}_t = 1\} \in \mathbb{Z}_{\geq 0} \cup \{\infty\}$ denote the first replacement time. For any $t \in \mathbb{Z}_{\geq 0}$ with $t \leq \tau_{\mathbf{r}}$, let $\mathbf{h}_t \in \mathcal{H}_t$ be the random variable denoting the first officeholder's public career history up to time t . For any $t \in \mathbb{Z}_{\geq 0}$, let $\pi_t := \pi(\mathbf{h}_{t \wedge \tau_{\mathbf{r}}}) \in [0, 1]$ denote the first officeholder's reputation, and let $\tilde{\mathbf{a}}_t := (1 - \mathbf{r}_t)\sigma_P(\mathbf{h}_{t \wedge \tau_{\mathbf{r}}}) \in [0, 1]$ denote the expected effort of the initial officeholder conditional on being opportunistic. Let $\mathbf{r}_\infty$ [resp., π_∞ , $\tilde{\mathbf{a}}_\infty$] denote the limit as $t \rightarrow \infty$ of $\mathbf{r}_t$ [resp., π_t , $\tilde{\mathbf{a}}_t$] in the event that this limit exists.

Any equilibrium induces a joint law over the stochastic process $(\mathbf{a}_t, \pi_t, \mathbf{r}_t)_{t=0}^\infty$. We take all probabilistic statements to be with respect to this law unless stated otherwise (and we suppress the underlying probability space and the specific equilibrium in our notation).

For ease of notation, we work throughout with the above random variables concerning only the first officeholder. Given our focus on personal symmetric equilibria, results we prove about the first officeholder apply without change to any officeholder.

We now show that politicians' incentives imply a lower bound on the opportunistic type's (time-averaged) hazard rate of being replaced.

Lemma 4. *Some $T \in \mathbb{Z}_{>0}$ and $\eta \in (0, 1)$ exist such that, in any equilibrium and at any history $h \in \mathcal{H}$ with $\sigma_P(h) > 0$, an opportunistic-type officeholder has a probability of at least η of being replaced in the next T periods.*

Proof. Take some large enough $T \in \mathbb{Z}_{>0}$ and small enough $\varepsilon \in (0, 1)$ to ensure that

$$(1 - \varepsilon)(1 - \delta^T) > (1 - \delta)(1 - \kappa) + \delta. \quad (\text{A.1})$$

Denoting $S_0 := \text{supp}(f_0)$, let $\lambda := \min_{s \in S_0} \frac{f_1(s)}{f_0(s)} \in (0, 1)$ and $\eta := \lambda^T \varepsilon$.

First, observe that at any history $h \in \mathcal{H}$ with $\sigma_P(h) > 0$, incentive compatibility requires that an opportunistic incumbent has a probability at least ε of being replaced in the next T periods, conditional on always shirking starting from h . The reason is that if this probability of replacement were strictly less than ε , then the continuation payoff from deviating to always shirking would exceed $(1 - \varepsilon)(1 - \delta^T)$, which by (A.1) strictly exceeds the maximum feasible continuation payoff from working at h .

Next, observe that for any sequence of T signals, its probability of arising (conditional on the incumbent not being replaced) under any politician strategy is at least λ^T times as high as its probability of arising if the politician were to always shirk. Therefore, the incumbent at h will be replaced in the next T periods with probability at least $\lambda^T \varepsilon = \eta$. $\square$

We now record a lemma that says either an officeholder's type will be revealed or learning will halt. The reasoning behind this lemma is essentially the same as that of [Cripps, Mailath, and Samuelson \(2004\)](#).

Lemma 5. *In any equilibrium, the stochastic process $(\mathbf{r}_t, \tilde{\mathbf{a}}_t, \boldsymbol{\pi}_t)_{t=0}^\infty$ satisfy the following:*

1. *The sequence $(\mathbf{r}_t, \boldsymbol{\pi}_t)_t$ converges a.s. to a random variable $(\mathbf{r}_\infty, \boldsymbol{\pi}_\infty)$.*
2. *We a.s. have $\mathbf{r}_\infty = 1$ (that is, $\boldsymbol{\tau}_\mathbf{r} < \infty$) or $\boldsymbol{\pi}_\infty \in \{0, 1\}$ or $(\tilde{\mathbf{a}}_t)_t$ converges to 1.*

Proof. The first point follows from pathwise monotonicity of $\mathbf{r}$ and the martingale convergence theorem for the bounded martingale $\boldsymbol{\pi}$.

For the second point, consider the event in which $\mathbf{r}_\infty \neq 1$ and $\boldsymbol{\pi}_\infty \notin \{0, 1\}$ —so $\mathbf{r}_\infty = 0$ (that is, $\boldsymbol{\tau}_\mathbf{r} = \infty$) and $0 < \boldsymbol{\pi}_\infty < 1$. For any $\pi \in (0, 1)$, we have $\min_{s \in S, \tilde{a} \in [0, 1]} f_{\pi + (1-\pi)\tilde{a}}(s) > 0$; and some $s \in S$ has $f_1(s) \neq f_0(s)$. Therefore, when a belief lies in some small enough

open neighborhood of π and the officeholder's expected effort is bounded away from 1, the probability of the belief exiting said neighborhood is bounded below. Thus, in the event that $0 < \pi_\infty < 1$, we must have $\tilde{\mathbf{a}}_t \rightarrow 1$. $\square$

We now prove that in any equilibrium, an officeholder is either eventually replaced or his type is asymptotically revealed.

Lemma 6. *Consider any equilibrium, and let $(\mathbf{r}_\infty, \pi_\infty)$ be as in the statement of Lemma 5. We a.s. have $\mathbf{r}_\infty = 1$ (that is, $\tau_{\mathbf{r}} < \infty$) or $\pi_\infty \in \{0, 1\}$.*

Proof. Consider the event in which $\pi_\infty > 0$ and $\tilde{\mathbf{a}}_\infty := \lim_{t \rightarrow \infty} \tilde{\mathbf{a}}_t = 1$. In this event, we in particular have $\tilde{\mathbf{a}}_t > 0$ for sufficiently large $t \in \mathbb{Z}_{\geq 0}$. Lemma 4 delivers some $T \in \mathbb{Z}_{>0}$ and $\eta \in (0, 1)$ such that the probability of replacement in the next T periods, conditional on the opportunistic-type officeholder, is eventually at least η in this event. But then, because $\tilde{\mathbf{a}}_\infty = 1$, the good type also eventually has at least a $\frac{\eta}{2}$ probability of replacement in the next T periods. Hence, by the Borel-Cantelli Lemma, with probability 1 conditional on this event, we have $\mathbf{r}_\infty = 1$.

We have argued that, excluding a null event, if $\pi_\infty > 0$ and $\tilde{\mathbf{a}}_\infty = 1$, then $\mathbf{r}_\infty = 1$. Combining this observation with Lemma 5 delivers the lemma. $\square$

The next lemma says that, when the replacement cost is small, an opportunistic officeholder is eventually replaced.

Lemma 7. *If $c < \pi_0$, then with probability 1 conditional on being an opportunistic type, an officeholder is replaced.*

Proof. Suppose $c < \pi_0$. Let us first show that (excepting a zero probability event) $\mathbf{r}_\infty = 1$ if $\pi_\infty = 0$. Indeed, if $\pi_\infty = 0$, then some $\tau \in \mathbb{Z}_{\geq 0}$ exists such that $\pi_t < \min\{c, \frac{1}{2}\}$ for every $t \geq \tau$. Let us now show $\mathbf{r}_\infty = 1$ in two different cases. First, if $\tilde{\mathbf{a}}_t = 0$ for some $t \geq \tau$, then voter incentives require $\mathbf{r}_t = 1$ so that $\mathbf{r}_\infty = 1$ too. Second, if $\tilde{\mathbf{a}}_t > 0$ for every $t \geq \tau$, then Lemma 4 delivers some $T \in \mathbb{Z}_{>0}$ and $\eta \in (0, 1)$ such that the probability of replacement in the next T periods is at least $\frac{\eta}{2}$ from time τ onward. Then, by the Borel-Cantelli Lemma, we have $\mathbf{r}_\infty = 1$.

Lemma 6 tells us $\mathbf{r}_\infty = 1$ or $\pi_\infty \in \{0, 1\}$ a.s., and the argument above rules out the case in which $\mathbf{r}_\infty = \pi_\infty = 0$ with positive probability. Therefore, we a.s. have $\mathbf{r}_\infty = 1$ or $\pi_\infty = 1$. The lemma follows because the probability of $\pi_\infty = 1$ conditional on the officeholder being an opportunistic type is necessarily zero. $\square$

The next lemma shows that, if an officeholder attains a high reputation on-path that the voter would currently retain him irrespective of opportunistic-type behavior, then he has a positive probability of never being replaced.

Lemma 8. *Fix an equilibrium, and let u_0 denote the voters' outside option. For any $t \in \mathbb{Z}_{\geq 0}$ and history $h \in \mathcal{H}$ is such that $\mathbb{P}\{\mathbf{h}_t = h \text{ and } \mathbf{r}_t = 0\} > 0$ and $\pi(h) > u_0 - c$, we have $\mathbb{P}\{\mathbf{h}_t = h, \pi_\infty \geq \pi(h), \text{ and } \mathbf{r}_\infty = 0\} > 0$.*

Proof. Define the stopping time $\tau := \inf\{\tau \in \mathbb{Z}_{\geq 0} : \tau \geq t \text{ and } \pi_\tau \leq u_0 - c\}$. Note that the random variable π_τ is well defined a.s. (even if $\tau = \infty$) by the martingale convergence theorem. Doob's optional stopping theorem implies $\mathbb{E}[\pi_\tau \mid \mathbf{h}_t = h] = \pi(h)$. Therefore, $\mathbb{P}\{\pi_\tau \geq \pi(h) \mid \mathbf{h}_t = h\} > 0$. But if $\mathbf{h}_t = h$ and $\pi_\tau \geq \pi(h) > u_0 - c$, then the voters' beliefs remain in $(u_0 - c, 1]$ from time t onward—and hence, by voter incentives, they do not replace the officeholder from time t onward. So

$$\mathbb{P}\{\mathbf{r}_\infty = 0 \text{ and } \pi_\infty \geq \pi(h) \mid \mathbf{h}_t = h \text{ and } \mathbf{r}_t = 0\} \geq \mathbb{P}\{\pi_\tau \geq \pi(h) \mid \mathbf{h}_t = h \text{ and } \mathbf{r}_t = 0\} > 0.$$

Because $\mathbb{P}\{\mathbf{h}_t = h \text{ and } \mathbf{r}_t = 0\} > 0$ by hypothesis, the result follows. $\square$

The following lemma says that, if a public career history is consistent with a officeholder not having been replaced yet, then some good news (that is, some signal indicative of effort) exists that does not lead to immediate firing.

Lemma 9. *In any equilibrium, any on-path history h admits some $s \in S$ with $f_1(s) > f_0(s)$ and (h, s) on path.*

Proof. Let $\mathcal{H}^*$ denote the set of on-path histories: $h \in \mathcal{H}$ if and only if some $t \in \mathbb{Z}_{\geq 0}$ has $\mathbb{P}\{\mathbf{h}_t = h \text{ and } \mathbf{r}_t = 0\} > 0$. First, consider the case in which $\sigma_P(h) = 0$. That $h \in \mathcal{H}^*$ while $\sigma_P(h) = 0$ implies $\pi(h) \geq u_0 - c$, where u_0 is the voters' outside option. Indeed, this inequality holds by voter incentives at h if $h \neq \emptyset$, and it holds by definition (since $c \geq 0$) for $h = \emptyset$. In this case, we can use any $s \in S$ with $f_1(s) > f_0(s)$ —which exists because f is informative. The Bayesian property implies (since $\pi(\cdot) > 0$ at on-path histories because f_1 has full support) that $\pi(h, s) > \pi(h)$. Hence $\pi(h, s) + [1 - \pi(h, s)]\sigma_P(h, s) > u_0 - c$, and so voters have a strict incentive to keep the incumbent at history (h, s) . Thus, $(h, s) \in \mathcal{H}^*$.

Now, turn to the case in which $\sigma_P(h) > 0$. The fact that the incumbent optimally chooses

to work at h implies

$$\begin{aligned}
0 &\leq \left\{ (1-\delta)(1-\kappa) + \delta \sum_{s \in S} f_1(s) [1 - \sigma_V(h, s)] v(h, s) \right\} \\
&\quad - \left\{ (1-\delta)1 + \delta \sum_{s \in S} f_0(s) [1 - \sigma_V(h, s)] v(h, s) \right\} \\
&= -(1-\delta)\kappa + \delta \sum_{s \in S} [f_1(s) - f_0(s)] [1 - \sigma_V(h, s)] v(h, s) \\
&< \sum_{s \in S: f_1(s) > f_0(s)} \delta [f_1(s) - f_0(s)] [1 - \sigma_V(h, s)] v(h, s).
\end{aligned}$$

Therefore, at least one summand is strictly positive. In particular, some $s \in S$ has $f_1(s) > f_0(s)$ and $\sigma_V(h, s) < 1$. Because $h \in \mathcal{H}^*$, the latter property tells us $(h, s) \in \mathcal{H}^*$ too. $\square$

The next lemma says that when [Condition FEI](#) fails, on-path equilibrium reputation can become arbitrarily optimistic.

Lemma 10. *Suppose [Condition FEI](#) fails, and fix any equilibrium. Any $\hat{\pi} \in (0, 1)$ has $\mathbb{P}\{\pi_t > \hat{\pi} \text{ and } \mathbf{r}_t = 0\} > 0$ for some $t \in \mathbb{Z}_{\geq 0}$.*

Proof. We begin with some useful definitions and notation. Let $\mathcal{H}^*$ be the set on-path histories (as also defined at the start of the proof of [Lemma 9](#)), let $\Pi^* := \{\pi(h) : h \in \mathcal{H}^*\}$, and let $\bar{\pi} := \sup(\Pi^*) \in [\pi_0, 1]$. Define $S_1 := \{s \in S : f_1(s) > f_0(s)\}$, and note that the function $(a, \pi) \mapsto \min_{s \in S_1} \beta_a(\pi|s)$ is continuous in on $[0, 1] \times (0, 1)$ because each of the finitely many maximands is. Hence, because $[0, 1]$ is compact and the function above takes $(1, \pi) \mapsto \pi$ for $\pi \in (0, 1)$, the function $\underline{a} : (0, 1) \rightarrow [0, 1]$ given by

$$\underline{a}(\pi) := \min \left\{ a \in [0, 1] : \min_{s \in S_1} \beta_a(\pi|s) \leq \bar{\pi} \right\}$$

is the lower envelope of a compact-valued and upper hemicontinuous correspondence—hence is well-defined and lower semicontinuous.

Our goal is to prove $\bar{\pi} = 1$; assume otherwise ($0 < \bar{\pi} < 1$) for a contradiction. Given this hypothesis, observe that any $a \in [0, 1]$ and $s \in S_1$ have $\beta_a(\bar{\pi}|s) > \bar{\pi}$; thus, $\underline{a}(\bar{\pi}) = 1$. This fact has two consequences. First, lower semicontinuity of $\underline{a}$ tells us $\lim_{\pi \rightarrow \bar{\pi}} \underline{a}(\pi) = 1$ too. Second, consequently, some $\underline{\pi}_0 \in (0, \bar{\pi})$ has $\underline{a}(\underline{\pi}_0) > 0$ and so $\min_{s \in S_1} \beta_0(\underline{\pi}_0|s) > \bar{\pi}$.

Now, let T be as given by [Lemma 2](#). Recursively for $t \in \{1, \dots, T\}$, let us find some $\underline{\pi}_t \in (\underline{\pi}_{t-1}, \bar{\pi})$ with the property that: Every $a \in [0, 1]$ with $\inf_{\pi \in (\underline{\pi}_t, \bar{\pi}), s \in S_1} \beta_a(\pi|s) \leq \bar{\pi}$ has

$\inf_{\pi \in (\underline{\pi}_t, \bar{\pi}), s \in S} \beta_a(\pi|s) \geq \underline{\pi}_{t-1}$. To do so, observe that $\lim_{\pi \rightarrow \bar{\pi}} \underline{a}(\pi) = 1$, and so (by continuity) every $s \in S$ has $\beta_{\underline{a}(\pi)}(\pi|s) \rightarrow \beta_1(\bar{\pi}|s) = \bar{\pi}$ as $\pi \rightarrow \bar{\pi}$. Hence, we can set $\underline{\pi}_t \in (\underline{\pi}_{t-1}, \bar{\pi})$ close enough to $\bar{\pi}$ to ensure $\beta_{\underline{a}(\underline{\pi}_t)}(\underline{\pi}_t|s) > \underline{\pi}_{t-1}$ for every $s \in S$. Let us see that such a $\underline{\pi}_t$ has the desired property. Indeed, any $a \in [0, 1]$ with $\inf_{\pi \in (\underline{\pi}_t, \bar{\pi}), s \in S_1} \beta_a(\pi|s) \leq \bar{\pi}$ will satisfy $\min_{s \in S_1} \beta_a(\underline{\pi}_t|s) \leq \bar{\pi}$ (by continuity and monotonicity of $\beta_a(\cdot|s)$), so that $a \geq \underline{a}(\underline{\pi}_t)$. Hence, any $s \in S$ satisfies $\beta_a(\underline{\pi}_t|s) \in \text{co}\{\beta_{\underline{a}(\underline{\pi}_t)}(\underline{\pi}_t|s), \beta_1(\underline{\pi}_t|s)\} = [\beta_{\underline{a}(\underline{\pi}_t)}(\underline{\pi}_t|s), \underline{\pi}_t]$. In particular, $\beta_a(\underline{\pi}_t|s) > \underline{\pi}_{t-1}$. Then, since s was arbitrary in this calculation and $\beta_a(\cdot|s)$ is increasing for every $a \in [0, 1]$ and $s \in S$, it follows that $\inf_{\pi \in (\underline{\pi}_t, \bar{\pi}), s \in S} \beta_a(\pi|s) > \underline{\pi}_{t-1}$.

Let us turn now to our equilibrium. Let $h_0 \in \mathcal{H}^*$ be some on-path history such that $\underline{\pi}_T < \pi(h_0) < \bar{\pi}$. Define $s_0, \dots, s_{T-1} \in S$ and $h_1, \dots, h_T \in \mathcal{H}$ as defined in the statement of [Lemma 3](#). That lemma tells us some $\tau \in \{0, \dots, T-1\}$ has $\sigma_P(h_\tau) = 0$; consider the smallest such τ . For every $t \in \{0, \dots, \tau-1\}$, the incumbent finds it optimal to exert effort at history h_t , which implies $[1 - \sigma_V(h_t, s)]v(h_t, s) > 0$ for some $s \in S$, hence for $s = s_t$. In particular, $\sigma_V(h_{t+1}) < 1$. By induction on t , it then follows that $h_1, \dots, h_\tau \in \mathcal{H}^*$.

Next, we show that $\underline{\pi}_{T-t} < \pi(h_t) < \bar{\pi}$ for every $t \in \{0, \dots, \tau\}$. We will prove this by induction; the base case of $t = 0$ holds by fiat. Toward the inductive step, suppose $t \in \{0, \dots, \tau-1\}$ has $\underline{\pi}_{T-t} < \pi(h_t) < \bar{\pi}$. Because $h_t \in \mathcal{H}^*$, [Lemma 9](#) tells us some $s'_t \in S_1$ has $(h_t, s'_t) \in \mathcal{H}^*$ too. But then, by definition of $\bar{\pi}$, it must be that $\pi(h_t, s'_t) \leq \bar{\pi}$. It then follows from the defining property of $\underline{\pi}_{T-t}$ that

$$\pi(h_{t+1}) = \pi(h_t, s_t) > \underline{\pi}_{T-t-1} = \underline{\pi}_{T-(t+1)},$$

completing the inductive step.

Finally, we now know that $h_\tau \in \mathcal{H}^*$, that $\pi(h_\tau) \in (\underline{\pi}_{T-\tau}, \bar{\pi}) \subseteq (\underline{\pi}_0, \bar{\pi})$ and that $\sigma_P(h_\tau) = 0$. [Lemma 9](#) says some $s \in S_1$ has $(h_\tau, s) \in \mathcal{H}^*$, and the definition of $\underline{\pi}_0$ means $\pi(h_\tau, s) > \bar{\pi}$, which contradicts the definition of $\bar{\pi}$. $\square$

Now we prove that when [Condition FEI](#) fails, any good-type officeholder has a positive probability of never being replaced.

Lemma 11. *Suppose [Condition FEI](#) fails. In any equilibrium, the probability of any given good-type officeholder never being replaced is strictly positive.*

Proof. Let us first show the result holds assuming the voters' outside option $u_0 \in [0, 1]$ satisfies $u_0 - c < 1$. Then we will verify this inequality holds.

By symmetry, we need only show the result for the first officeholder. By [Lemma 10](#) (and since $u_0 - c < 1$), there is a positive probability of reaching a history at which the first

officeholder has not been replaced, and her reputation is strictly positive and strictly greater than $u_0 - c$. But then [Lemma 8](#) says the probability of never being replaced and having $\pi_\infty > 0$ is strictly positive too. Hence, the probability $\mathbb{E}[(1 - \mathbf{r}_\infty)\pi_\infty]$ of being a good type and never being replaced is strictly positive.

All that remains is to show $u_0 - c < 1$. The inequality is trivial if either $u_0 < 1$ or $c > 0$, so assume towards contradiction that $u_0 = 1$ and $c = 0$. Voter incentives then require $\sigma_P(h) = 1$ at any history $h \in \mathcal{H}$ with $\pi(h) < 1$ and $\sigma_V(h) < 1$. Because the Bayesian property then implies $\pi(h) = \pi_0 < 1$ at any history that is reached with positive probability, the opportunistic-type officeholder works at every on-path history. But this contradicts politician incentives, by [Lemma 1](#). $\square$

Now we argue that when [Condition FEI](#) fails and replacement costs are low, a good-type politician is eventually retained in office forever.

Proof of [Proposition 3](#). The first part (which relies on $c < \pi_0$) is exactly [Lemma 7](#), and the second (which does not require our bound on replacement cost) is [Lemma 11](#).

Now, toward the proposition's last sentence (which again relies on $c < \pi_0$), suppose [Condition FEI](#) fails. We know every opportunistic officeholder is replaced with probability 1, whereas every good-type officeholder has a positive (and by symmetry, the same positive) probability of never being replaced. By the Borel-Cantelli Lemma, with probability 1, some officeholder is never replaced, and this officeholder is a good type. $\square$

A.2.4. Equilibrium Constructions

The next lemma says [Condition FEI](#) is equivalent to a more stringent condition that imposes further restrictions on the continuation values. In terms of induced behavior, if full effort can be incentivized at all, it can be done with a stationary likelihood-ratio test in which sufficiently bad news results in certain replacement.

Lemma 12. *If [Condition FEI](#) holds, then it can be witnessed by $v \in \mathbb{R}_{\geq 0}^S$ such that:*

- *Some $\bar{v} \in (0, 1 - \kappa)$ exists such that $v(S) = \{v(s) : s \in S\} = \{0, \bar{v}\}$.*
- *Every s_0 with $f_0(s_0) \leq f_1(s_0)$ has $v(s_0) = \bar{v}$.*
- *Every $s_0, s_1 \in S$ with $v(s_0) = 0$ and $v(s_1) = \bar{v}$ has $\frac{f_0(s_0)}{f_1(s_0)} > \frac{f_0(s_1)}{f_1(s_1)}$.*
- *Inequality (PK_{FEI}) holds with equality.*

Proof. Let $S_0 := \{s \in S : f_1(s) < f_0(s)\}$, and let $\succsim$ be the complete transitive binary relation on S given by $s' \succsim s \iff \frac{f_0(s')}{f_1(s')} \leq \frac{f_0(s)}{f_1(s)}$. Suppose [Condition FEI](#) holds and is

witnessed by v . We will successively show that various properties of v are without loss of generality. To that end, let $\bar{v} := \max v(S) \in \mathbb{R}_{\geq 0}$. Note that $0 < \bar{v} < 1 - \kappa$, since any v with all entries the same would violate (IC_{FEI}) , any v with $\max v(S) \geq 1 - \kappa$ would violate (PK_{FEI}) .

First, raising $v(s)$ to $\bar{v}$ for each $s \in S \setminus S_0$ relaxes both constraints. So we may assume $v(s) = \bar{v}$ for every $s \in S \setminus S_0$. Second, if $\hat{S} \subseteq S_0$ is some $\sim$ -equivalence class, then replacing $v(s)$ with $\frac{\sum_{\tilde{s} \in \hat{S}} f_1(\tilde{s})v(\tilde{s})}{\sum_{\tilde{s} \in \hat{S}} f_1(\tilde{s})}$ for every $s \in \hat{S}$ preserves both constraints. So we may assume v is constant on every $\sim$ -equivalence class.

Next, observe that—because $v \in [0, \bar{v}]^S$ —some $s^* \in S_0$ exists such that

$$\sum_{s \succ s^*} f_1(s) \bar{v} \leq \sum_{s \in S} f_1(s) v(s) \leq \sum_{s \prec s^*} f_1(s) \bar{v};$$

and so some $\hat{v} \in [0, \bar{v}]$ exists such that

$$\left[\sum_{s \succ s^*} f_1(s) \right] \bar{v} + \left[\sum_{s \sim s^*} f_1(s) \right] \hat{v} = \sum_{s \in S} f_1(s) v(s).$$

Observe now that both inequalities in [Condition FEI](#) will still be satisfied if we replace v with the vector whose s entry is $\bar{v}$ for $s \succ s^*$, is $\hat{v} \in [0, \bar{v}]$ for $s \sim s^*$, and is 0 for $s \prec s^*$. Indeed, this modification has no effect on (PK_{FEI}) but relaxes (IC_{FEI}) . (Moreover, because we had $v|_{S \setminus S_0} = v$ before the rearrangement and $S \setminus S_0 \succ S_0$, it follows that we still have this property.) So we may further assume v takes this form.

Now, define $\hat{z} := 1 - \delta \sum_{\tilde{s} \succ s^*} f_1(\tilde{s}) > 0$ and $\bar{z} := -\delta \sum_{\tilde{s} \sim s^*} f_1(\tilde{s}) < 0$, and define $z \in \mathbb{R}^S$ by letting

$$z(s) := \begin{cases} 0 & : s \prec s^* \\ \hat{z} & : s \sim s^* \\ \bar{z} & : s \succ s^* \end{cases}$$

for each $s \in S$. Observe that the set of $\varepsilon \in \mathbb{R}$ with $\bar{v} + \varepsilon \bar{z} \geq \hat{v} + \varepsilon \hat{z} \geq 0$ is a closed interval containing zero. It is also bounded: the first inequality is violated as $\varepsilon \rightarrow \infty$, and the second is violated as $\varepsilon \rightarrow -\infty$. This interval therefore has two extreme points, and at least one of these ε is such that $v + \varepsilon z$ satisfies (IC_{FEI}) because the latter is affine. Finally, simple algebra shows that this $v + \varepsilon z$ satisfies (PK_{FEI}) too because v does. So replacing v with this modification, we may assume that $\hat{v} \in \{0, \bar{v}\}$, or equivalently that $v(S) \subseteq \{0, \bar{v}\}$. And in fact, we have $v(S) = \{0, \bar{v}\}$ because any v with all entries the same violates (IC_{FEI}) .

Finally, because scaling up a vector v that satisfies (IC_{FEI}) relaxes this constraint but tightens (PK_{FEI}) , we can scale it up to ensure (PK_{FEI}) holds with inequality. $\square$

We now prove that under [Condition FEI](#) and a small replacement cost, some equilibrium attains full effort.

Lemma 13. *If $c \leq 1 - \pi_0$ and [Condition FEI](#) holds, then some equilibrium attains full effort.*

Proof. By [Lemma 12](#), some $\bar{v} \in (0, 1)$ and $\lambda \geq 1$ exist such that $S^* := \left\{ s \in S : \frac{f_0(s)}{f_1(s)} \leq \lambda \right\}$ satisfies

$$\bar{v} = (1 - \delta)(1 - \kappa) + \delta \sum_{s \in S^*} f_1(s) \bar{v} \geq (1 - \delta)1 + \delta \sum_{s \in S^*} f_0(s) \bar{v}.$$

Let $\mathcal{H}_{\text{Pass}} := \bigcup_{t=0}^{\infty} (S^*)^t$, the histories in which the incumbent has never generated a signal outside of S^* . We will now construct an equilibrium that attains full effort. Let the politician strategy be given by $\sigma_P(h) = \mathbf{1}_{h \in \mathcal{H}_{\text{Pass}}}$, the voters' strategies be given by $\sigma_V(h) = \mathbf{1}_{h \notin \mathcal{H}_{\text{Pass}}}$, and let the belief map be given by

$$\pi(h) = \begin{cases} \pi_0 & \text{if } h \in \mathcal{H}_{\text{Pass}} \\ \pi_0 & \text{if } h = (\tilde{h}, s) \text{ for some } \tilde{h} \in \mathcal{H}_{\text{Pass}} \text{ and } s \in S \setminus S^* \\ 0 & \text{otherwise.} \end{cases}$$

This strategy profile clearly leads to effort always being chosen until an incumbent is replaced, so we need only show this construction is in fact an equilibrium.²⁸

A incumbent's incentives at histories outside of $\mathcal{H}_{\text{Pass}}$ are satisfied because he will be replaced immediately after any signal, and so it is optimal to shirk. Meanwhile, his incentives at histories in $\mathcal{H}_{\text{Pass}}$ follow directly from the properties defining $\bar{v}$: his continuation value at such a history is $\bar{v}$, and he prefers this to the value of a one-shot deviation in which he shirks. A voter's incentives at histories inside $\mathcal{H}_{\text{Pass}}$ are satisfied because she receives effort with probability 1 whatever she does, so she prefers not to bear the nonnegative replacement cost. For her incentives at histories $h \notin \mathcal{H}_{\text{Pass}}$, her payoff from not replacing is $\pi(h) \leq \pi_0$, and her payoff from replacing is $1 - c \geq \pi_0$. Finally, the Bayesian property is straightforward: $\pi(\emptyset) = \pi_0$, and at every history h with $\sigma_V(h) < 1$, we have $\sigma_P(h) = 1$ and $\pi(h, s) = \pi(h)$ for every $s \in S$. $\square$

²⁸ The construction is “nearly” Markovian: play depends only on beliefs and the most recent signal. In fact, because beliefs do not change on path, the equilibrium is “simple” in the sense of [Banks and Sundaram \(1993\)](#). Our construction in the proof of [Lemma 14](#) is also nearly Markovian, but because beliefs do change on path it is not simple in their sense.

The next lemma shows how to construct an equilibrium that does not attain eventual full effort, given [Condition FEI](#) and a small replacement cost. This equilibrium has every incumbent being eventually replaced and a positive probability of shirking from new incumbents.

Lemma 14. *If $c < 1 - \pi_0$ and [Condition FEI](#) holds, then some equilibrium does not attain eventual full effort.*

Proof. We begin with some useful notation and calculations. First, by [Lemma 12](#), some $\bar{v} \in (0, 1 - \kappa)$ and $\lambda \in \left[1, \max_{s \in S} \frac{f_0(s)}{f_1(s)}\right)$ exist such that, letting

$$S^* := \left\{s \in S : \frac{f_0(s)}{f_1(s)} \leq \lambda\right\}, \text{ and } f_a^* := \sum_{s \in S^*} f_a(s) \in [0, 1) \text{ for } a \in \{0, 1\},$$

we have $\bar{v} = (1 - \delta)(1 - \kappa) + \delta f_1^* \bar{v} \geq (1 - \delta)1 + \delta f_0^* \bar{v}$. The previous inequality tells us $f_1^* > f_0^*$ so that $\tilde{v} := \bar{v} - \frac{1 - \delta}{\delta} \frac{\kappa}{f_1^* - f_0^*} < \bar{v}$. Observe also that

$$\begin{aligned} \delta(f_1^* - f_0^*)\tilde{v} &= \delta(f_1^* - f_0^*)\bar{v} - (1 - \delta)\kappa \\ &= [(1 - \delta)(1 - \kappa) + \delta f_1^* \bar{v}] - [(1 - \delta)1 + \delta f_0^* \bar{v}] \\ &\geq 0, \end{aligned}$$

so that $\tilde{v} \geq 0$. Now, defining $\hat{v} := (1 - \delta)(1 - \kappa) + \delta[f_1^* \bar{v} + (1 - f_1^*)\tilde{v}]$, note that $\hat{v}$ is a proper weighted average of the three terms $\tilde{v} < \bar{v} < 1 - \kappa$, and so is strictly higher than $\tilde{v}$. Having established that $0 \leq \tilde{v} < \hat{v}$, we thus know that $x := 1 - \frac{\tilde{v}}{\hat{v}} \in (0, 1]$. Moreover,

$$\begin{aligned} &(1 - \delta) + \delta[f_0^* \bar{v} + (1 - f_0^*)\tilde{v}] - \hat{v} \\ &= (1 - \delta)[1 - (1 - \kappa)] + \delta\{[f_0^* \bar{v} + (1 - f_0^*)\tilde{v}] - [f_1^* \bar{v} + (1 - f_1^*)\tilde{v}]\} \\ &= (1 - \delta)\kappa + \delta(f_0^* - f_1^*)(\bar{v} - \tilde{v}) \\ &= 0. \end{aligned}$$

In summary, given the above calculations and substituting in $\tilde{v} = (1 - x)\hat{v}$, we have that

$$\begin{aligned}
x &\in (0, 1], \\
\bar{v} &= (1 - \delta)(1 - \kappa) + \delta f_1^* \bar{v} \\
&\geq (1 - \delta)1 + \delta f_0^* \bar{v}, \text{ and} \\
\hat{v} &= (1 - \delta)(1 - \kappa) + \delta [f_1^* \bar{v} + (1 - f_1^*)(1 - x)\hat{v}] \\
&= (1 - \delta) + \delta [f_0^* \bar{v} + (1 - f_0^*)(1 - x)\hat{v}].
\end{aligned}$$

We will use these calculations to construct an equilibrium of the desired form.

To construct such an equilibrium, fix some $a_0 \in \left(\frac{c}{1 - \pi_0}, 1\right)$, which exists by hypothesis. We will next describe a strategy profile and belief map that depends on a_0 (though we will suppress this dependence in our notation). We will then argue that the profile does not generate EFE, and that for a_0 close enough to 1 it is an equilibrium.

To describe the strategy profile and belief map, partition histories into three categories.²⁹ Say a history is a *first-regime* history if no signal in S^* has ever (with this incumbent) appeared; say a history is a *second-regime* history if it is not a first-regime history, and every signal since the first S^* signal has been in S^* ; and say a history is a *third-regime* history otherwise. Consider strategies and beliefs $(\sigma_P, \sigma_V, \pi)$ as follows:

- In the first regime: Both voter and incumbent mix. Specifically, the voter always chooses $\sigma_V(h) = x$. We recursively construct politician behavior and voter beliefs at a history h in this regime as follows. Start by setting $\sigma_P(\emptyset) = a_0$ and $\pi(\emptyset) = \pi_0$. Then consider any $h = (h_-, s)$ in this regime with $h_- \in \mathcal{H}$ and $s \in S$. Given $\sigma_P(h_-) > 0$, which holds by induction, set $\pi(h) = \beta_{\sigma_P(h_-)}(\pi(h_-)|s) \in (0, \pi(h_-))$, as signal s is strictly bad news (recall $\lambda \geq 1$ and $s \notin S^*$). Set $\sigma_P(h) \in [a_0, 1)$ so that $\pi(h) + [1 - \pi(h)]\sigma_P(h) = \pi_0 + (1 - \pi_0)a_0 - c$; doing so is possible because the definition of a_0 ensures $\pi_0 + (1 - \pi_0)a_0 - c > \pi_0 \geq \pi(h)$.
- In the second regime: The voter retains the incumbent, and the incumbent works. That is, the voter always chooses $\sigma_V(h) = 0$, and the politician always chooses $\sigma_P(h) = 1$. Write $h = (h_-, s)$ for some $h_- \in \mathcal{H}$ and $s \in S$. As $\sigma_P(h_-) > 0$ by construction, set $\pi(h) = \beta_{\sigma_P(h_-)}(\pi(h_-)|s)$.
- In the third regime: The voter replaces the incumbent, and the incumbent shirks. That is, the voter always chooses $\sigma_V(h) = 1$, and the politician always chooses $\sigma_P(h) = 0$.

²⁹ Recall that [Figure 1](#) depicts the strategy profile for the special case of binary signals.

Write $h = (h_-, s)$ for some $h_- \in \mathcal{H}$ and $s \in S$. If h_- is a second-regime history, then $\sigma_P(h_-) = 1$, so set $\pi(h) = \beta_{\sigma_P(h_-)}(\pi(h_-)|s) = \pi(h_-)$. If h_- is not a second-regime history, then $\sigma_V(h_-) = 1$ by construction, so we may set $\pi(h) = 0$.³⁰

Below, we will argue that the constructed $(\sigma_P, \sigma_V, \pi)$ is an equilibrium for appropriate choice of the parameter a_0 . Before doing so, we point out that the lemma would follow if it is. Specifically, EFE fails because every officeholder shirks with positive probability at his initial history, and almost surely every officeholder is eventually replaced. To confirm the latter point, note that (i) if an officeholder is not replaced at a first-regime history, then there is a probability bounded away from zero (at least $[\pi_0 + (1 - \pi_0)a_0 - c]f_1^* > \pi_0 f_1^*$) of transitioning to the second regime in the next period; (ii) from the second regime, there is a constant positive probability of transitioning to the third regime in the next period; and (iii) the officeholder is replaced at every third-regime history. So the Borel-Cantelli lemma implies almost sure eventual replacement of any officeholder.

We now confirm that $(\sigma_P, \sigma_V, \pi)$ is an equilibrium for some a_0 . The Bayesian property is immediate from the construction. Next, consider politician incentives. These are trivial in the third regime: it is optimal for him to shirk since he will be replaced immediately no matter which signal he generates. His incentives to work in the second regime follow directly from the definition of $\bar{v}$ and S^* . For the first regime, we can use the fact that

$$\hat{v} = (1 - \delta)(1 - \kappa) + \delta [f_1^* \bar{v} + (1 - f_1^*)(1 - x)\hat{v}] = (1 - \delta) + \delta [f_0^* \bar{v} + (1 - f_0^*)(1 - x)\hat{v}].$$

These equalities tell us that his continuation value from any first-regime history is $\hat{v}$, and that he is indifferent between his two effort choices there. Now, we turn to voter incentives. In the second regime, the voter gets payoff 1 from keeping the incumbent, and so does so optimally. In the first regime, the voter is indifferent because $\pi(h) + [1 - \pi(h)]\sigma_P(h) = \pi_0 + (1 - \pi_0)a_0 - c$.

All that remains is to verify voter optimality in the third regime. Because the voter's outside option is $\pi_0 + (1 - \pi_0)a_0$ and the opportunistic type always shirks in the third regime, it is optimal for the voter to replace at a third-regime history h if and only if

$$\pi(h) \leq \pi_0 + (1 - \pi_0)a_0 - c. \tag{A.2}$$

Toward that inequality, let us show that some $\bar{s} \in S$ has

$$\pi(h) \leq \beta_{a_0}(\pi_0|\bar{s}). \tag{A.3}$$

³⁰ Here is the only other place (besides that noted in [footnote 26](#)) where we use the flexibility of off-path beliefs afforded by *weak* PBE.

First, if the previous history was not a second-regime history, then $\pi(h) = 0$ and so (A.3) holds for any $\bar{s} \in S$. So consider the complementary case, in which $h = (h_2, s_2)$ for some second-regime history h_2 and some $s_2 \in S$. By definition of the second regime, there exists some first-regime history h_1 (which comprises only signals from $S \setminus S^*$), some signal $s_1 \in S^*$, and some (possibly empty) string h_+ of signals from S^* such that $h_2 = (h_1, s_1, h_+)$. Every history $\tilde{h}$ between³¹ (h_1, s_1) and h_2 has $\sigma_P(\tilde{h}) = 1$, and so $\pi(\tilde{h}, \tilde{s}) = \pi(\tilde{h})$ for every signal $\tilde{s}$, implying $\pi(h) = \pi(h_2) = \pi(h_1, s_1)$. Since every $s \in S \setminus S^*$ has $f_1(s) < f_0(s)$, it follows that $0 < \beta_a(\pi|s) \leq \pi$ for every $\pi \in (0, 1]$ and $a \in [0, 1]$; hence, by induction $\pi(h_1) \leq \pi(\emptyset) = \pi_0$. Hence, by construction, the first-regime history h_1 has $\sigma_P(h_1) \geq \sigma_P(\emptyset) = a_0$. Therefore:

- If $f_1(s_1) > f_0(s_1)$, we have $\pi(h_1, s_1) = \beta_{\sigma_P(h_1)}(\pi(h_1) | s_1) \leq \beta_{a_0}(\pi(h_1) | s_1) \leq \beta_{a_0}(\pi_0 | s_1)$. So (A.3) holds for $\bar{s} = s_1$.
- If $f_1(s_1) \leq f_0(s_1)$, we have $\pi(h_1, s_1) = \beta_{\sigma_P(h_1)}(\pi(h_1) | s_1) \leq \pi(h_1) \leq \pi_0$. So (A.3) holds for any $\bar{s} \in S$ with $f_1(\bar{s}) \geq f_0(\bar{s})$.

Given that (A.3) holds for some $\bar{s} \in S$, inequality (A.2) follows if

$$\max_{s \in S} \beta_{a_0}(\pi_0 | s) \leq \pi_0 + (1 - \pi_0)a_0 - c.$$

This inequality holds when a_0 is close enough to 1 because as $a_0 \rightarrow 1$, the right-hand side converges to $1 - c$ while the left-hand side converges to $\pi_0 < 1 - c$. $\square$

A.2.5. The Equivalence Theorem

Proof of Theorem 1. Recall that the theorem assumes our maintained assumption in the main text that $c < \min\{\pi_0, 1 - \pi_0\}$.

First, suppose Condition FEI fails. Proposition 3 then tells us that every equilibrium attains eventual full effort. Yet no equilibrium attains full effort because politician incentive-compatibility would be violated (Lemma 1).

Next, suppose Condition FEI holds. Lemma 13 then implies that some equilibrium attains full effort; and Lemma 14 demonstrates that not every equilibrium attains eventual full effort. $\square$

A.3. Other Results

³¹ That is, for any history $\tilde{h}$ that weakly succeeds (h_1, s_1) and weakly precedes h_2 .

A.3.1. Replacement Despite a Favorable Reputation

Proof of Proposition 2. Consider any equilibrium. We want to show it either attains eventual full effort or has $\mathbb{P}\{\pi_{\tau_r} > \pi_0 \text{ and } \mathbf{r}_\infty = 1\} > 0$. We prove the result in three cases.

First, if $\mathbb{P}\{\tau_r = \infty\} > 0$, then the Borel-Cantelli Lemma and symmetry of the equilibrium tell us that, with probability 1, some officeholder is never replaced. But since (by Lemma 7) that officeholder is a good type with conditional probability 1, the equilibrium attains eventual full effort. Second, if $\mathbb{P}\{\pi_{\tau_r} = \pi_0\} = 1$, then (because $f_1 \neq f_0$ and beliefs are a martingale) signals are uninformative of effort on path, so the equilibrium attains full effort, hence eventual full effort.

Consider now the remaining case in which $\tau_r < \infty$ a.s. and $\mathbb{P}\{\pi_{\tau_r} = \pi_0\} < 1$. Then, Doob's optional stopping theorem implies $\mathbb{E}[\pi_{\tau_r}] = \pi_0$. Hence, $\mathbb{P}\{\pi_{\tau_r} > \pi_0\} > 0$, delivering the result. $\square$

A.3.2. Voter Welfare in the Long Run

Proof of Corollary 1. Suppose Condition FEI fails, and that $c < \pi_0$. Fix an equilibrium, and let B_t denote the event that the time- t incumbent is a commitment type who will never be replaced. Observe $B_0 \subseteq B_1 \subseteq \dots$ by definition, and Proposition 3 tells us $\mathbb{P}(\bigcup_{t=0}^{\infty} B_t) = 1$. Continuity along chains then implies $\mathbb{P}(B_t) \rightarrow 1$ as $t \rightarrow \infty$. But the time- t voter's ex-ante expected payoff is in $[\mathbb{P}(B_t), 1]$, delivering the corollary. $\square$

A.3.3. Voters' Outside Option

The following lemma bounds the growth on incumbent reputation, and hence the growth rate of a voter's value from keeping the incumbent, for a given lower bound on the opportunistic officeholder's expected effort.

Lemma 15. *For any $t \in \mathbb{Z}_{\geq 0}$ and $\eta \in (0, 1)$, the map $\bar{\beta}_\eta^t$ is increasing, and any $\pi \in (0, 1)$ has*

$$\bar{\beta}_\eta^t(\pi) + [1 - \bar{\beta}_\eta^t(\pi)] \eta \leq \frac{\pi + (1 - \pi)\eta^{t+1}}{\pi + (1 - \pi)\eta^t}.$$

Moreover, the right-hand side of the above inequality is increasing in t .

Proof. In the expression defining $\beta_a(\pi|s)$, the numerator does not depend on a , whereas the denominator is affine (hence monotone) in it, so that $\beta_a(\pi|s)$ is monotone in a too. Therefore,

$$\min_{a \in [\eta, 1]} \beta_a(\pi|s) = \max\{\beta_\eta(\pi|s), \beta_1(\pi|s)\} = \max\{\beta_\eta(\pi|s), \pi\}.$$

For each $s \in S$, observe that

$$\beta_\eta(\pi|s) = \frac{\pi}{\pi + (1 - \pi) \left[\eta + (1 - \eta) \frac{f_0(s)}{f_1(s)} \right]},$$

which is strictly decreasing in $\frac{f_0(s)}{f_1(s)}$. Letting $\lambda := \min_{s \in S} \frac{f_0(s)}{f_1(s)} \in [0, 1]$, it follows that

$$\bar{\beta}_\eta(\pi) = \max_{s \in S} \max \{ \beta_\eta(\pi|s), \pi \} = \max \left\{ \max_{s \in S} \beta_\eta(\pi|s), \pi \right\} = \frac{\pi}{\pi + (1 - \pi) [\eta + (1 - \eta) \lambda]} \in (0, 1).$$

This quantity is increasing in π because $\eta + (1 - \eta) \lambda \leq 1$, and so the composition $\bar{\beta}_\eta^t$ is also increasing.

We now pursue the upper bound. That $\lambda \geq 0$ tells us $\bar{\beta}_\eta(\pi) \leq \frac{\pi}{\pi + (1 - \pi) \eta}$, implying

$$\frac{\bar{\beta}_\eta(\pi)}{1 - \bar{\beta}_\eta(\pi)} \leq \frac{\pi}{(1 - \pi) \eta}.$$

Because $\bar{\beta}_\eta$ is increasing, it follows from induction that $\frac{\bar{\beta}_\eta^t(\pi)}{1 - \bar{\beta}_\eta^t(\pi)} \leq \frac{\pi}{(1 - \pi) \eta^t}$ for every $t \in \mathbb{Z}_{\geq 0}$. Rearranging this inequality yields $\bar{\beta}_\eta^t(\pi) \leq \frac{\pi}{\pi + (1 - \pi) \eta^t}$, and so (since $\eta \leq 1$)

$$\begin{aligned} \bar{\beta}_\eta^t(\pi) + [1 - \bar{\beta}_\eta^t(\pi)] \eta &\leq \frac{\pi}{\pi + (1 - \pi) \eta^t} + \frac{(1 - \pi) \eta^t}{\pi + (1 - \pi) \eta^t} \eta \\ &= \frac{\pi + (1 - \pi) \eta^{t+1}}{\pi + (1 - \pi) \eta^t} \in (0, 1). \end{aligned}$$

All that remains now is to see that this bound increases with t . And indeed, the bound is equal to

$$\frac{\frac{\pi}{1 - \pi} + \eta \cdot \eta^t}{\frac{\pi}{1 - \pi} + \eta^t},$$

which (because $0 < \eta < 1$) is strictly decreasing in η^t and so strictly increasing in t . $\square$

The next lemma derives a parametric upper bound on the voters' outside option.

Lemma 16. *Suppose [Condition FEI](#) fails, and let T be as delivered by [Lemma 2](#). In any equilibrium, voters' outside option is bounded above by*

$$c + \inf_{\eta \in (0, 1)} \frac{\pi_0 + (1 - \pi_0) \eta^{T+1}}{\pi_0 + (1 - \pi_0) \eta^T}.$$

Proof. Fixing an equilibrium, assume for a contradiction that the voters' outside option u_0

has $u_0 - c > \frac{\pi_0 + (1 - \pi_0)\eta^{T+1}}{\pi_0 + (1 - \pi_0)\eta^T}$ for some $\eta \in (0, 1)$.

Let $h_0 := \emptyset$, and then let $s_0, \dots, s_{T-1} \in S$ and $h_1, \dots, h_T \in \mathcal{H}$ be as in [Lemma 3](#).

Now, let us show by induction that every $t \in \{0, \dots, T-1\}$ has both $\sigma_P(h_t) \geq \eta$ and $\pi(h_t) \leq \bar{\beta}_\eta^t(\pi_0)$. For the base case ($t = 0$), observe that $\bar{\beta}_\eta^0(\pi_0) = \pi_0 = \pi(h_0)$, while

$$\pi_0 + (1 - \pi_0)\sigma_P(h_0) = u_0 > c + \frac{\pi_0 + (1 - \pi_0)\eta^T \eta}{\pi_0 + (1 - \pi_0)\eta^T} \geq 0 + \frac{\pi_0 + (1 - \pi_0)\eta \cdot 1}{\pi_0 + (1 - \pi_0)1} = \pi_0 + (1 - \pi_0)\eta \cdot 1,$$

so that $\sigma_P(h_0) \geq \eta$.

Let us turn next to the inductive step. Suppose $t \in \{0, \dots, T-2\}$ has both $\sigma_P(h_t) \geq \eta$ and $\pi(h_t) \leq \bar{\beta}_\eta^t(\pi_0)$; we aim to show $t+1$ enjoys the same properties. First, we have

$$\pi(h_{t+1}) \leq \bar{\beta}_\eta(\pi(h_t)) \leq \bar{\beta}_\eta(\bar{\beta}_\eta^t(\pi_0)) = \bar{\beta}_\eta^{t+1}(\pi_0),$$

where the first inequality holds because $\sigma_P(h_t) \geq \eta$, and the second inequality holds (given [Lemma 15](#)) because $\pi(h_t) \leq \bar{\beta}_\eta^t(\pi_0)$. Second, observe that $\sigma_P(h_t) \geq \eta > 0$ would not be incentive compatible if $[1 - \sigma_V(h_t, s)]v(h_t, s)$ were zero for every $s \in S$. Hence, given the definition of s_t and h_{t+1} , we necessarily have $\sigma_V(h_{t+1}) < 1$. Voter incentives then imply

$$\begin{aligned} u_0 - c &\leq \pi(h_{t+1}) + [1 - \pi(h_{t+1})]\sigma_P(h_{t+1}) \\ &\leq \bar{\beta}_\eta^{t+1}(\pi_0) + [1 - \bar{\beta}_\eta^{t+1}(\pi_0)]\sigma_P(h_{t+1}). \end{aligned}$$

Meanwhile, by hypothesis and given [Lemma 15](#), we have

$$\begin{aligned} u_0 - c &> \frac{\pi_0 + (1 - \pi_0)\eta^{T+1}}{\pi_0 + (1 - \pi_0)\eta^T} \\ &\geq \bar{\beta}_\eta^{t+1}(\pi_0) + [1 - \bar{\beta}_\eta^{t+1}(\pi_0)]\eta. \end{aligned}$$

Combining the two inequality chains yields $\sigma_P(h_{t+1}) \geq \eta$, completing the inductive step.

That $\sigma_P(h_t) > 0$ for every $t \in \{0, \dots, T-1\}$ contradicts [Lemma 3](#). $\square$

To prove [Proposition 4](#), we show that the bound of [Lemma 16](#) can be made arbitrarily small if the prior and replacement costs are both small.³²

³² Although [Section 2](#) assumes $c < \min\{\pi_0, 1 - \pi_0\}$, the restriction is evidently not needed for [Proposition 4](#): if [Condition FEI](#) fails and c and π_0 are both very small, then the outside option is very small in all equilibria, irrespective of the relationship between c and π_0 .

Proof of Proposition 4. Let T be as given by Lemma 2. Because

$$\begin{aligned} 0 &\leq c + \inf_{\eta \in (0,1)} \frac{\pi_0 + (1 - \pi_0)\eta^{T+1}}{\pi_0 + (1 - \pi_0)\eta^T} \\ &\leq c + \frac{\pi_0 + (1 - \pi_0)\pi_0^{(T+1)/(T+2)}}{\pi_0 + (1 - \pi_0)\pi_0^{T/(T+2)}} \\ &\rightarrow 0 \text{ as } (c, \pi_0) \rightarrow (0, 0), \end{aligned}$$

it follows that $c + \inf_{\eta \in (0,1)} \frac{\pi_0 + (1 - \pi_0)\eta^{T+1}}{\pi_0 + (1 - \pi_0)\eta^T}$ converges to zero as c and π_0 do. The proposition then follows from Lemma 16. $\square$

Finally, although not stated in the main text, we record one further result—Corollary 2 below—that extends our main equivalence result. It uses the following lemma that converts a violation of Condition FEI into an upper bound on the voters’ outside option.

Lemma 17. *For any $(\kappa, \delta, f, \pi_0)$ such that (κ, δ, f) do not satisfy Condition FEI, some $\bar{c} > 0$ exists such that the supremum voter outside option from any equilibrium with replacement cost $c \in [0, \bar{c}]$ is strictly below 1.*

Proof. Let T be as given by Lemma 2. In light of Lemma 16, we could take any $\bar{c} > 0$ with $c + \inf_{\eta \in (0,1)} \frac{\pi_0 + (1 - \pi_0)\eta^{T+1}}{\pi_0 + (1 - \pi_0)\eta^T} < 1$. And indeed, such a $\bar{c}$ exists because $\frac{\pi_0 + (1 - \pi_0)\eta^{T+1}}{\pi_0 + (1 - \pi_0)\eta^T} < 1$ for any given $\eta \in (0, 1)$. $\square$

The following corollary extends the equivalences in Theorem 1 to the voters’ outside option being bounded away from first best, so long as the replacement cost is sufficiently small.

Corollary 2. *For any $(\kappa, \delta, f, \pi_0)$, there exists $\bar{c} > 0$ such that, if $c < \bar{c}$, then the following are equivalent:*

1. *All equilibria attain eventual full effort;*
2. *No equilibrium attains full effort;*
3. *Condition FEI fails.*
4. *Across all equilibria, the voters’ outside option is bounded away from 1.*

Proof. Let $\hat{c} := \min\{\pi_0, 1 - \pi_0\}$. If (κ, δ, f) satisfy Condition FEI, then we can let $\bar{c} := \hat{c}$. If (κ, δ, f) violate Condition FEI, then we can take $\bar{c}$ to the minimum of $\hat{c}$ and the bound delivered by Lemma 17. Theorem 1 then tells us conditions 1, 2, and 3 are all equivalent; and obviously condition 4 implies 2. Lastly, Lemma 17 tells us condition 3 implies 4. $\square$

Symbol	Description	Defined In
<i>Histories, Strategies, and Values</i>		
$\mathcal{H}$	Set of all (public) career histories	§2
$f_{\tilde{a}}(s)$	Probability of signal s given $\tilde{a}$ expected effort: equals $\tilde{a}f_1(s) + (1 - \tilde{a})f_0(s)$	§2
$\sigma_P(h)$	Probability an opportunistic incumbent exerts effort at history h	§2
$\sigma_V(h)$	Probability the voter replaces the incumbent at history h	§2
$\pi(h)$	Incumbent's reputation (belief that he is the good type) at history h	§2
$\alpha(h)$	Expected effort at history h : equals $\pi(h) + [1 - \pi(h)]\sigma_P(h)$	§2
$v(h)$	Opportunistic incumbent's continuation value at history h	§2
u_0	Voters' outside option: equals $\pi_0 + (1 - \pi_0)\sigma_P(\emptyset)$	§3.4
<i>Belief Operators</i>		
$\beta_a(\pi \mid s)$	Bayesian posterior from belief π , opportunistic type's effort a , and signal s	§A.2.2
$\bar{\beta}_\eta(\pi)$	Highest possible Bayesian posterior from belief π and opportunistic type's effort at least η	§A.2.2
$\bar{\beta}_\eta^t$	The t -fold composition of the operator $\bar{\beta}_\eta$	§A.2.2
<i>Random Variables</i>		
$\mathbf{r}_t$	Indicator variable for replacement: equals 1 if first ^a officeholder has been replaced by time t	§A.2.3
$\tau_{\mathbf{r}}$	Replacement time of first officeholder: equals $\inf\{t : \mathbf{r}_t = 1\} \in \mathbb{Z}_{\geq 0} \cup \{\infty\}$	§A.2.3
π_t	First officeholder's reputation at time t (stopped at $\tau_{\mathbf{r}}$)	§A.2.3
$\tilde{\mathbf{a}}_t$	Officeholder's expected effort at time t conditional on being opportunistic	§A.2.3
$\pi_\infty, \mathbf{r}_\infty, \tilde{\mathbf{a}}_\infty$	Almost-sure limits of the stochastic processes $\pi_t, \mathbf{r}_t, \tilde{\mathbf{a}}_t$	§A.2.3

^a By equilibrium symmetry, an analog applies to other officeholders for this and other relevant random variables

Table 2: Summary of Key Notation used in the Proof Appendix

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